



Dynamics of intramammary infections and their association with quarter-level somatic cell score, milk yield, and milk components. Part 1: Staphylococcal species

Daryna Kurban,^{1,2,3} Jean-Philippe Roy,^{2,3,4} Trevor J. DeVries,^{3,5} Greg P. Keefe,^{3,6} Pamela R. F. Adkins,⁷ John R. Middleton,⁷ and Simon Dufour^{1,2,3*}

¹Department of Pathology and Microbiology, Faculty of Veterinary Medicine, Université de Montréal, Saint-Hyacinthe, QC, Canada J2S 2M2

²Regroupement FRQNT Op+lait, Saint-Hyacinthe, QC, Canada J2S 2M2

³Mastitis Network, Saint-Hyacinthe, QC, Canada J2S 2M2

⁴Department of Clinical Sciences, Faculty of Veterinary Medicine, Université de Montréal, Saint-Hyacinthe, QC, Canada J2S 2M2

⁵Department of Animal Sciences, University of Guelph, ON, Canada N1G 2W1

⁶Department of Health Management, Atlantic Veterinary College, University of Prince Edward Island, Charlottetown, PE, Canada C1A 4P3

⁷Department of Veterinary Medicine and Surgery, University of Missouri, Columbia, MO 65211

ABSTRACT

On Canadian dairy farms, staphylococcal species, including *Staphylococcus aureus* and NAS, cause most IMI in lactating cows. However, knowledge remains limited about their impact on quarter-level SCS, milk yield, and composition, as well as their lactational incidence and elimination rates. This study aimed to describe staphylococcal species-specific IMI association with SCS, milk production, and milk composition in primiparous and multiparous cows, and to describe their prevalence, incidence, and elimination rates. Five commercial dairy farms with automated milking systems in Québec, Canada, were visited biweekly to collect aseptic quarter milk samples. Cows were recruited at 3 to 14 DIM and sampled every other week until 305 DIM. Bacteriological culture and MALDI-TOF MS were used for staphylococcal identification. A total of 21,467 samples from 1,537 quarters of 387 cows (307 multiparous, 80 primiparous) were analyzed. The most prevalent staphylococci in mammary gland quarters were *Staphylococcus chromogenes* (4.0%), *Staphylococcus epidermidis* (2.6%), *Staphylococcus haemolyticus* (2.1%), and *S. aureus* (1.9%). *Staphylococcus auricularis*, *Staphylococcus chromogenes*, and *Staphylococcus equorum* IMI were more frequent in primiparous cows, whereas *S. aureus*, *Staphylococcus epidermidis*, and *S. haemolyticus* were more common in multiparous cows. *Staphylococcus haemolyticus* and *S. epidermidis* showed the highest incidence, whereas *S. chromogenes* and *Staphylococcus simulans* had elimination rates similar to or lower than

S. aureus. Many lactational staphylococcal IMI were transient, with high species-specific elimination rates. Associations with quarter-level SCS, milk yield, and milk components were evaluated using 7,703 to 9,967 test-day samples from over 1,300 quarters of 369 cows. In primiparous cows, quarters with IMI by *S. aureus*, *S. chromogenes*, *S. epidermidis*, or *S. simulans* had higher SCS than healthy quarters. In multiparous cows, SCS was elevated in quarters infected by *S. aureus*, *S. chromogenes*, *S. epidermidis*, *S. haemolyticus*, *S. simulans*, or *Staphylococcus xylosus*. Milk yields in primiparous cows were generally unaffected by IMI status, whereas in multiparous cows, only *S. aureus* IMI reduced milk yield (7.9 vs. 10.2 kg/d in healthy quarters). No changes in milk fat were detected. Protein content was higher in *S. aureus* IMI quarters of multiparous cows. Lactose reductions occurred in primiparous cows' quarters with *S. chromogenes* or *S. simulans* IMI, and in multiparous cows' quarters with *S. aureus*, *S. epidermidis*, or *S. simulans* IMI. In conclusion, in primiparous cows, IMI by *S. aureus*, *S. chromogenes*, *S. epidermidis*, and *S. simulans* increased SCS, with *S. simulans* also reducing lactose, but no changes in milk yield, fat, or protein. In multiparous cows, higher SCS occurred with several staphylococcal species, milk yield was reduced only with *S. aureus* IMI, and lactose was reduced in quarters infected by *S. aureus*, *S. epidermidis*, and *S. simulans*.

Key words: *Staphylococcus*, incidence rate, elimination rate, somatic cell score, milk production and components

INTRODUCTION

Mastitis is responsible for substantial economic losses in dairy farms, mainly because of milk quality and production reduction, early culling of cows, and increased

Received October 14, 2025.

Accepted March 18, 2026.

*Corresponding author: simon.dufour@umontreal.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

veterinary and treatment costs (Aghamohammadi et al., 2018). Intramammary infections are the primary cause of bovine subclinical and clinical mastitis. Intramammary infections are characterized by the presence of pathogenic microorganisms in milk, which induce an elevation in inflammatory cells and, sometimes, visible inflammation of the mammary gland tissue (Ruegg, 2023).

Based on phenotypic and pathogenic properties, staphylococcal species are often classified into *Staphylococcus aureus* and NAS. Although *S. aureus* is well recognized as one of the major pathogens causing both clinical and subclinical mastitis, NAS are often considered as minor pathogens causing only subclinical mastitis and, rarely, mild clinical mastitis (Reyher et al., 2012; Ruegg, 2017). However, NAS are represented by more than 50 different species; thus, the species-specific epidemiology and udder health impact may vary among them (Vanderhaeghen et al., 2015). In particular, a recent review highlighted that many NAS, such as *Staphylococcus chromogenes*, *Staphylococcus agnetis*, *Staphylococcus arlettae*, *Staphylococcus auricularis*, *Staphylococcus capitis*, *Staphylococcus devriesei*, *Staphylococcus equorum*, *Staphylococcus gallinarum*, *Staphylococcus haemolyticus*, *Staphylococcus sciuri*, *Staphylococcus simulans*, *Staphylococcus hominis*, *Staphylococcus intermedius*, *Staphylococcus microti*, and *Staphylococcus succinus*, are associated not only with high SCC, but also with clinical mastitis (CM) cases (Kurban et al., 2022).

In Canada, staphylococcal species (*S. aureus* and NAS) are the most frequent group of microorganisms isolated from milk and are responsible for the majority of IMI in lactating dairy cows. Relative IMI prevalence in apparently normal milking quarters of 2.5% for *S. aureus* and 42.7% for NAS were reported in Canada (Dufour et al., 2012a,b; Condas et al., 2017). In CM cases, *S. aureus* was recovered from 13% of samples, whereas NAS were the most commonly isolated group of pathogens from nonclinical lactating cows (Reyher et al., 2011). Species-specific NAS prevalence varies among dairy herds and different geographical locations (De Buck et al., 2021). However, the most frequently recovered and identified species are usually *S. chromogenes*, *S. simulans*, *Staphylococcus xylosus*, *S. haemolyticus*, and *Staphylococcus epidermidis* (De Buck et al., 2021; Kurban et al., 2022).

Beyond their high prevalence, NAS are also very dynamic regarding IMI acquisition and elimination. For NAS IMI, an incidence rate of 0.29 new IMI per quarter-month and an elimination rate of 0.79 eliminated IMI per infected quarter-month have been reported (Dufour et al., 2012a). However, except for *S. aureus* IMI, limited data are available on species-specific incidence and elimination rates of staphylococcal IMI during the lactating period. Some NAS IMI (e.g., *S. chromogenes*, *S. simulans*,

S. xylosus) are known to be persistent (Fry et al., 2014) and have been associated with a moderate increase of SCC, with a geometric mean of up to 442,000 cells/mL, thus comparable to *S. aureus* IMI (Supré et al., 2011; De Visscher et al., 2016).

Although *S. aureus* IMI cause milk quality alterations and milk yield losses (Coulon et al., 2002; Djabri et al., 2002; Botaro et al., 2015), the effect of NAS IMI on milk production is still debatable. Some researchers have not associated NAS IMI, as a group, with milk yield changes (Hertl et al., 2014; Tomazi et al., 2015). However, others have described a lower (Taponen and Pyorala, 2009; Heikkilä et al., 2018) or higher (Piepers et al., 2013) milk yield in infected cows.

Regarding the impact on milk composition, Botaro et al. (2015) reported no effect of *S. aureus* subclinical mastitis on milk protein or lactose, despite an overall decrease in milk production and fat yield. Nevertheless, Chen et al. (2021) demonstrated that intracellular *S. aureus* infections reduce milk protein synthesis by decreasing amino acids uptake in bovine mammary epithelial cells. In comparison to *S. aureus* IMI, Tomazi et al. (2015) reported no milk component changes in quarters infected by NAS, when evaluated either at the group or species levels.

To our knowledge, no scientific literature is available on the association of species-specific bovine-associated staphylococcal IMI with changes in quarter-level milk components (fat, protein, and lactose) during the whole lactation. Little data are available on associations of NAS species-specific IMI with changes in quarter SCC and quarter milk production during the lactating period. Moreover, most of the previous investigations of relationship of NAS IMI and changes in SCC, milk yield, or milk composition faced the following limitations: (1) NAS were mainly studied as a group due to the complexity of obtaining a precise species-level identification, thus disregarding species-specific differences; (2) SCC, milk yield, and milk composition were measured only at the cow level, whereas IMI generally affect individual infected quarters; and (3) SCC, milk production, and milk composition data were measured once or for a limited period, whereas some IMI caused by NAS might persist during the lactation. Finally, although some data are available on the association of species-specific NAS IMI and changes in SCC and milk production in heifers, we cannot extrapolate these results to multiparous cows due to physiological and lactational differences. Nevertheless, knowledge on the expected changes in milk yield or composition associated with a given NAS IMI could provide valuable information for decision-making.

Therefore, our main objective was to report the association of staphylococcal species-specific IMI with possible alterations in quarter SCS, quarter milk production,

and milk components (fat, protein, and lactose) during lactation and by parity (primiparous vs. multiparous). Secondary objectives were to (1) describe staphylococcal IMI species-specific quarter-level apparent prevalence during the lactation period and (2) describe staphylococcal IMI species-specific quarter-level incidence and elimination rates during the lactation period.

MATERIALS AND METHODS

Study Design

This research was part of a longitudinal prospective observational cohort study on staphylococcal IMI. Sampling and laboratory analysis organization were described in (Kurban et al., 2025). Briefly, a convenience sample of 5 commercial dairy farms located within a 25-km radius of the Faculté de médecine vétérinaire of the Université de Montréal (Saint-Hyacinthe, QC, Canada) was selected. The selection of the farms allowed us to have a possibility to easily obtain 500 calvings (100 primiparous and 400 multiparous) within 18 mo. Based on the projected sample size and on estimates from Condas et al. (2017), when designing the study, we anticipated that in multiparous cows, between ~100 IMI (for the least prevalent species) and 1,000 IMI (for the most prevalent species) would be identified, to be compared with roughly 10,000 healthy quarter observations. In primiparous cows, between 30 and 300 IMI and ~3,000 healthy observations were expected. Assuming an SD of 2.0 kg for quarter-level daily milk yield, the expected number of IMI would provide 90% power (2-sided $\alpha = 0.05$) to detect a difference in quarter milk yield of ≥ 0.2 kg/d for the most prevalent species and ≥ 0.7 kg/d for the least prevalent species in multiparous cows. In primiparous cows, the corresponding detectable differences would be ≥ 0.4 kg/d and ≥ 1.2 kg/d, respectively, based on a 2-sample *t*-test. These calculations did not account for clustering of repeated observations within quarters and cows; therefore, the effective sample size is likely smaller than assumed, and the true statistical power is expected to be lower than 90%.

All herds recruited were clients of the Université de Montréal bovine ambulatory clinic and were using Lely Astronaut A4 or later generation (Lely Industries N.V., Maassluis, the Netherlands) automated milking systems (AMS). This selection criterion was required to later extract quarter-level milk production data. During our first visit, the research project was explained in detail to the producer, and a consent form was completed and signed. The study was approved by the Animal Use Ethics Committee of the Université de Montréal (protocol 18-Rech-1975).

Cow Selection and Milk Sampling

The sampling period lasted 14 mo, from December 2018 to March 2020. The data collection was completed in mid-March 2020 due to COVID-19 pandemic lockdown. Farms were visited every other week for the recruitment of cows and to collect samples. Holstein cows were recruited systematically at calving. Other breeds were excluded. On a given cow, the first quarter milk samples were collected starting at 3 to 14 DIM, and then every other week up to 305 DIM (to reach ~20 samples per quarter per lactation).

When necessary, cows were gently restrained to allow safe and hygienic sample collection. Restraint methods depended on the farm's facilities and included securing the cow in a resting stall using a rope positioned behind the animal or using headlocks in the feeding area when available. Approximately 50 mL of milk was collected aseptically from each quarter of each enrolled cow, according to the Mastitis Network recommendations for milk sample collection technique for bacterial testing (<http://www.reseaumammite.org/tactic/en/milk-sample-collection-technique-for-bacterial-testing/>). Briefly, the technical team wore gloves, cleaned the teats of bedding residues, and forestripped milk to stimulate the milk let-down and discarded the first milk streams. Teats were then disinfected by dipping them in the same disinfectant solution used in the AMS system on the farm, dried with disposable paper towels, and disinfected again with an alcohol swab until the swab remained clean. Without allowing the tube to come in contact with the teat end, ~50 mL of milk from the quarter was drawn into an individual aseptic vial. Samples were temporarily labeled using the cow and quarter number and later labeled with a unique barcode. The exact time of sampling was also recorded on the vials to be able to later correlate the timing of sample collection relative to when each cow was milked by the AMS. Finally, all teats were disinfected by dipping them in the farm's teat-dip solution immediately after sample collection.

Samples were placed on ice and transported on the same day to our research laboratory. As soon as samples arrived at our research laboratory, a 1-mL aliquot, aseptically transferred using a single-channel pipette and sterile tips for every milk sample, was frozen at -20°C for storage before microbiological analyses. A preservative containing bronopol and natamycin (Broad-Spectrum Microtabs II, Advanced Instruments, Norwood, MA) was added to the remaining milk, and the next day, these samples were sent to a milk quality laboratory (Lactanet, Sainte-Anne-de-Bellevue, QC, Canada) for SCC and milk composition analyses using a CombiFoss 7 DC instrument (FOSS, Hillerød, Denmark).

Somatic cell count results were transformed to SCS using the formula below (Wiggans and Shook, 1987, Čitek et al., 2022):

$$\text{SCS} = \log_2 \left(\frac{\text{SCC}}{100,000} \right) + 3,$$

where SCC is given in cells per milliliter of milk.

Quarter-level milk production data for every milking were extracted from the AMS software. The extracted files initially contained data on each visit of each cow to the AMS unit, with 2 cow-level identifiers for each cow, date and time of visit, milking visit information (successful milking, failure, or refusal), and quarter milk yield (in kg/visit), as well as other information (e.g., quarter-level milking time, quarter-level milk conductivity, cow-level milk yield). Data from cows enrolled during the current study and for the lactation of interest were identified and extracted. Subsequently, only information on AMS visits where a successful milking occurred was included. Using these data, for each DIM between 3 and 305 DIM, quarter milk yield from the preceding 72 h was summed up and divided by 3 to provide, for every day, a rolling daily 3-d average quarter milk production (in kg/d; Deluyker et al., 1990).

Milk Bacteriological Culture and Interpretation

Microorganisms in milk were identified using the same methodology as in Kurban et al. (2025). Briefly, bacteriological culture was followed by speciation using the MALDI-TOF method. Bacteriological culture was conducted following NMC guidelines (NMC, 2017). Milk samples were thawed, and a volume of 0.01 mL of each quarter milk sample was inoculated on blood agar (Columbia agar 5% sheep blood) and incubated at 35°C for 24 h and up to 48 h (when no growth was observed after 24 h) under aerobic conditions. Based on culture results, milk samples were classified as (1) pure culture (1 phenotype of colony-forming units), (2) mixed culture (2 different phenotypes of colony-forming units), (3) contaminated sample (≥ 3 different phenotypes of colony-forming units) and (4) no growth (when no growth was observed after 48 h of incubation). All contaminated samples were considered noninformative.

Then, for pure and mixed growth cultures, 1 cfu from each phenotypically different colony was identified by MALDI-TOF MS analysis using a MALDI Biotyper system (Bruker Daltonics Inc., Billerica, MA). Briefly, a sterile toothpick was used to collect material from each phenotypically different colony-forming unit (1 toothpick per colony-forming unit type), and the bacterial material was applied to a disposable ceramic plate with 96 targets

(1 target per isolate). Direct extraction of the isolate was carried out using an MBT Galaxy System device (Bruker Daltonics Inc., Billerica, MA). First, this device applied formic acid. After drying, an α -cyano-4-hydroxycinnamic acid matrix was applied, and the targets were dried again. The MALDI Biotyper system (Bruker Daltonics Inc., Billerica, MA) was then used to acquire the mass spectrum profile of each isolate on the plate. This profile was then compared with a database of reference spectra to identify the isolate. The MALDI-TOF manufacturer's reference database (MBT 8468 MSP Library) was enhanced with a custom-validated reference spectra database for NAS isolates from cattle to increase the proportion of species-level identification of staphylococci (Cameron et al., 2017). The manufacturer's suggested identification score (score < 1.7 : no identification possible; score of 1.7–2.0: identification up to genus level is valid; score of ≥ 2.0 : identification up to the species level is valid) was applied for interpreting the results. However, species-level identification of NAS identified with a score ≥ 1.7 was considered valid, as suggested by Cameron et al. (2018). Participating farms were not informed of the bacteriological culture results until the end of the study.

IMI Definition

All quarter milk samples with missing milk bacteriological culture results or interpreted as contaminated were considered uninformative and classified as missing data for subsequent analyses. Milk samples with pure or mixed culture results were considered to be positive for a microorganism identified as relevant for udder health whenever ≥ 100 cfu/mL (i.e., ≥ 1 cfu on the on the blood agar plate) of that microorganism was isolated from a milk sample. Only the staphylococcal species (the objective of this research) and the other 61 microorganism species (Supplemental Table S1, see Notes) reported by NMC (2016) and by Kurban et al. (2022) as mastitis pathogens were considered relevant.

Lactational Prevalence, Incidence, and Elimination Rates of Staphylococcal Species IMI. Quarter IMI status on the sampling day was interpreted based on bacteriological culture results of a single sample from this quarter (Dohoo et al., 2011a). This interpretation was chosen to identify as many IMI as possible, providing a better sensitivity and a lower proportion of false-negative results at the cost, however, of a lower specificity (Dohoo et al., 2011b).

Next, we jointly evaluated the quarter IMI status of 2 consecutive measurements to establish changes in IMI status. A quarter was considered to have acquired a new staphylococcal IMI by a given species if the first IMI status of the pair of results was negative for an IMI by a given staphylococcal species and the following IMI

status was positive for that species. Alternatively, the elimination of an IMI was deemed to have occurred when a quarter had a positive IMI status for a given staphylococcal species and then a negative status for that species on the next measurement. Whenever one or both samples of a pair were uninformative (missing bacteriological result or contaminated sample), then change in the quarter IMI status was considered unknown and the pair of samples was considered uninterpretable. Finally, whenever a change of IMI status was noted (i.e., acquisition of a new staphylococcal IMI or elimination of an existing staphylococcal IMI), it was deemed to have occurred immediately after the first of the 2 measurements (i.e., on the day that the first sample was collected).

Association of Staphylococcal IMI During Lactation with Changes in SCS, Milk Production, and Components (Fat, Protein, and Lactose). To report the effect of having a lactational staphylococcal IMI on SCS, milk production, and components, the quarter IMI status was interpreted based on evaluation of bacteriological culture results of 2 consecutive milk samples from this quarter. The quarter was assumed to have an IMI when both consecutive milk samples collected within 14 d were bacteriologically positive for the same staphylococcal species of interest. Likewise, the quarter was interpreted as healthy when bacteriological culture results of 2 consecutive milk samples from this quarter were negative to any staphylococcal or nonstaphylococcal microorganism. When a discordance was observed between the samples of a given pair, then the quarter status was deemed as unknown. This approach was chosen to provide a higher certainty, when computing effect of an IMI on milk yield and composition, that quarters identified as infected by a given species were truly infected, and that quarters identified as healthy were truly healthy quarters. As a sensitivity analysis, we also repeated the SCS, milk yield, and composition analyses using a more liberal quarter IMI status definition based on evaluation of a single milk sample.

Statistical Analyses

All statistical analyses were conducted using R (version 4.2.1; <https://www.R-project.org/>) within RStudio (version 2024.04.1; <https://posit.co/>). To make our results reproducible, the datasets and the R scripts used for analyses are made available as supplemental material (see Notes).

Lactational Prevalence of Staphylococcal Species IMI. The apparent prevalence of IMI by a given staphylococcal species was estimated for each of the most frequent staphylococcal species (i.e., >50 IMI identified in the dataset; Supplemental Table S2, see Notes). First, we calculated the overall prevalence of each staphylococcal

species using a logistic regression model without any predictors and using, as an outcome, whether the quarter had a positive IMI status to a given staphylococcal species on a given day (i.e., one model per staphylococcal species). This model, therefore, estimated an intercept and its 95% CI, which could then be transformed into an apparent prevalence using the inverse logit function. Due to estimation problems, robust variance (R sandwich package version 3.1.0; <https://CRAN.R-project.org/package=sandwich>) was used, rather than generalized mixed models, to account for the clustering of quarter IMI statuses by quarter, cow, and herd when estimating the IMI prevalence and its 95% CI.

We then computed the IMI apparent prevalence for each staphylococcal species as a function of DIM and parity (primiparous vs. multiparous), using the model described previously. To achieve this, the main terms and an interaction term between parity (primiparous vs. multiparous) and DIM (as a third-degree polynomial term) were added to the model as predictors. If the Wald statistic tests on the parity $\times$ DIM interaction terms all yielded P -values >0.05 , then we assumed that the IMI apparent prevalence did not vary as a function of parity during the production cycle. In such cases, the parity $\times$ DIM interaction terms were removed from the model. After removing the interaction terms, if the Wald statistic tests on the parity main term yielded P -values >0.05 , then we assumed that the IMI apparent prevalence did not vary as a function of parity, and the parity term was also removed from the model. Finally, LSM estimates and their 95% CI were computed (R emmeans package version 1.10.2; <https://CRAN.R-project.org/package=emmeans>), by 7-d steps, between 7 and 305 DIM in order to describe IMI prevalence as a function of DIM and, when relevant, as a function of parity.

Incidence and Elimination Rates of Staphylococcal Species IMI. The IMI incidence and elimination rates of each of the most frequent staphylococcal species were also estimated solely for the most frequent staphylococcal species (i.e., >50 IMI identified in the dataset). For this, we used the same modeling approach as the one described for IMI prevalence. However, to consider the time period between samples, which may vary, for instance when a sample was contaminated (time between 2 IMI statuses would then be >14 d), the number of days at risk (i.e., the number of days between 2 informative samples) was used as a weight variable in our logistic model. Moreover, only quarters at risk of experiencing a new IMI by a given staphylococcal species (i.e., quarters not infected by the staphylococcal species on the first measurement of a pair) were considered for the incidence analyses. Similarly, only quarters at risk of eliminating a given IMI species (i.e., quarters infected by the staphylococcal species on the first measurement

of a pair) were considered for the elimination analyses. With this modeling approach, applying the inverse logit function on the obtained estimates would then represent the proportion of quarters having experienced a new IMI (or having eliminated an existing IMI) among quarters at risk of experiencing this event and over the modal period at risk, (i.e., an IMI incidence or an IMI elimination rate per quarter per 14 d). Then, to express these incidence and elimination rates in new and eliminated IMI per 100 quarter-weeks, respectively, the obtained rates were divided by 2 and multiplied by 100.

Association of Staphylococcal IMI During Lactation with Changes in SCS, Milk Production, and Components (Fat, Protein, and Lactose). Finally, we quantified the impact of being infected with a given staphylococcal species on the quarter milk SCS, the quarter milk yield, and its fat, protein, and lactose composition. For these investigations, we only kept daily observations with an IMI status that was either negative to all microorganisms (defined as “healthy”) or positive to a given bacterial species, but in pure culture only. Thus, daily observations with mixed IMI were excluded because it would be impossible to attribute the SCS or milk yield variations to the presence of a specific microorganism. Within this dataset, only IMI by the most frequent staphylococcal species were considered; all the other staphylococcal species IMI were grouped together with the other microbial species IMI in an “other IMI” category (Supplemental Tables S2 and S3, see Notes). With this approach, we were able to obtain a variable classifying the quarter-day IMI status as healthy versus *S. aureus*, versus *S. auricularis*, versus *S. chromogenes*, versus *S. epidermidis*, versus *S. equorum*, versus *S. haemolyticus*, versus *S. hominis*, versus *S. simulans*, versus *S. xylosus*, versus other IMI.

To describe the overall association of the quarter IMI status with quarter milk SCS, production, fat, protein, and lactose we used mixed linear models with either the sampling day quarter milk results for SCS, quarter milk yield, or milk components (fat, protein, and lactose separately) as outcomes and, as predictors, the quarter-day IMI status and its interaction with parity (primiparous vs. multiparous). Within this model, we accounted for the repeated observations within a quarter using a random intercept with a repeated measures correlation structure. More precisely, a spatial exponential correlation structure was used when modeling quarter milk SCS, fat, protein, and lactose (thus, accounting for potentially nonequidistant measures), and an autoregressive first-order moving average structure was used for daily quarter milk yield (because this measure was always available every day, i.e., equidistant data). Moreover, cow and herd random intercepts were used to account for the clustering of quarters by cows and of cows by herd, respectively. These analyses were conducted using the R nlme package

(version 3.1-157; <https://CRAN.R-project.org/package=nlme>). The model was as follows:

$$Y_{ijkl} = \beta_0 + \beta_1 \text{IMI status}_{ijkl} + \beta_2 \text{Parity}_{kl} + \beta_3 \text{IMI status}_{ijkl} \times \text{Parity}_{kl} + h_i + c_{kl} + q_{jkl} + e_{ijkl},$$

where Y_{ijkl} is the quarter milk SCS, fat, protein, lactose or quarter daily milk yield (in kg/d) of the i th test-day from the j th quarter from the k th cow from the l th herd; β_0 , β_1 , β_2 , and β_3 are the intercept and regression coefficients; IMI status $_{ijkl}$ is the IMI status (e.g., healthy vs. *S. aureus*, vs. *S. chromogenes*, vs. *S. epidermidis*, vs. *S. haemolyticus*, vs. *S. simulans*, vs. *S. xylosus*, vs. other IMI) for the i th test-day of the j th quarter from the k th cow from the l th herd; Parity $_{kl}$ is the parity (primiparous vs. multiparous) of the k th cow from the l th herd; IMI status $_{ijkl} \times$ Parity $_{kl}$ is the interaction term between IMI status and parity; and h_i , c_{kl} , q_{jkl} , and e_{ijkl} are the herd, cow, quarter, and observation residuals, respectively.

Using this model, we were able to report the expected SCS, fat, protein, lactose concentrations, and milk yield of quarters from primiparous or multiparous cows, and classified them as either healthy or infected with a given staphylococcal species, averaged across all stages of lactation. To achieve this, we simply computed the LSM and their 95% CI. For these latter analyses, the Tukey method was used to account for multiple comparisons.

Then, to model the association of the quarter IMI status with changes in SCS, fat, protein, lactose, and milk yield, but now as a function of DIM, we used the same linear mixed model but included, as predictors, the quarter-day IMI status variable, parity, and DIM (as a 3-degree polynomial term to provide a smooth estimation of the SCS and lactation curves), and all the 2-way and the 3-way interaction terms. The LSM estimates and their 95% CI were then computed (R emmeans package version 1.10.2), by 7-d steps, between 7 and 305 DIM to describe the effect of the quarter IMI status on the quarter SCS or milk yield and as a function of DIM and parity.

Model Diagnostics. Homoscedasticity and normality assumptions for the linear mixed models were evaluated by plotting the standardized residuals against the predicted values and plotting the quantile plot, respectively. These plots could be generated for the observation-, cow- and herd-level residuals, but not for the quarter-level residuals because of the repeated measures correlation structure used for the latter.

RESULTS

Dairy Herd Characteristics

The characteristics of the 5 farms participating in our study are described in Table 1. The herd size ranged from

Table 1. Characteristics of the 5 commercial dairy farms (Québec, Canada) using an automated milking system (AMS), recruited for a study on the association of staphylococcal IMI with quarter-level milk SCS, milk production, and milk components (fat, protein, lactose)

Item	Farm 1	Farm 2	Farm 3	Farm 4	Farm 5
Herd size, number of lactating cows	150	60	95	380	210
Number of AMS units	3	1	2	8	4
Number of lactating cow pens	1	1	1	4 (parity 1, 2, 3, and ≥4)	2 (parity 1 and 2 together and parity ≥3)
Mean quarter milk SCS	2.2 (SD = 2.2)	2.5 (SD = 2.3)	2.3 (SD = 2.3)	2.9 (SD = 2.5)	2.5 (SD = 2.3)
Mean cow-level milk production, kg/cow per day	44.7 (SD = 10.1)	41.3 (SD = 10.3)	35.5 (SD = 10.1)	38.8 (SD = 10.8)	36.7 (SD = 10.5)
Mean quarter-level milk production, kg/quarter per day	11.2 (SD = 3.2)	10.1 (SD = 3.1)	8.7 (SD = 2.8)	9.4 (SD = 3.6)	9.3 (SD = 3.4)
Mean quarter-level fat in milk, %	4.7 (SD = 2.3)	4.8 (SD = 2.4)	5.1 (SD = 2.4)	5.1 (SD = 2.4)	4.7 (SD = 2.2)
Mean quarter-level protein in milk, %	3.2 (SD = 0.3)	3.1 (SD = 0.3)	3.3 (SD = 0.4)	3.2 (SD = 0.3)	3.2 (SD = 0.3)
Mean quarter-level lactose in milk, %	4.4 (SD = 0.3)	4.4 (SD = 0.3)	4.4 (SD = 0.3)	4.4 (SD = 0.3)	4.5 (SD = 0.3)
Bedding type: Lactating cows	Peat moss and straw mix on stall mats	Deep stalls with sand bedding	Padded mattress with wood shavings	Padded mattress with wood shavings	Padded mattress with wood shavings
Bedding type: Dry cows	Peat moss and straw mix + straw before calving	Straw	Straw	Wood shavings on mattress + straw before calving	Straw
Teat disinfection used	Iodine solution	Iodine solution	Iodine solution	Lactic acid-based foam product	Iodine solution
Dry-off protocol	Intramammary cephalixin, ¹ vaccine against <i>Escherichia coli</i> , ² <i>Salmonella Typhimurium</i> , ² and <i>Klebsiella pneumoniae</i> ³ to all cows	Intramammary cephalixin benzathine ¹ to all cows	Intramammary novobiocin and penicillin, ⁴ vaccine against <i>Escherichia coli</i> ² and <i>Salmonella Typhimurium</i> ² to all cows	Intramammary novobiocin and penicillin ⁴ and vaccine against <i>Escherichia coli</i> ² and <i>Salmonella Typhimurium</i> ² to all cows	Ceftiofur ⁵ to cows having a clinical mastitis before dry-off
Mean dry period duration (d)	61 (SD = 7)	58 (SD = 4)	55 (SD = 12)	56 (SD = 7)	54 (SD = 9)

¹Cefa-Dri (Boehringer-Ingelheim Animal Health Canada Inc., Burlington, ON, Canada). Active ingredient: cephalixin benzathine.²J-Vac (Boehringer-Ingelheim Animal Health Canada Inc., Burlington, ON, Canada). A broad-spectrum bacterin-toxoid vaccine that protects against coliform mastitis caused by *Escherichia coli* and the effects of endotoxemia caused by *E. coli* and *Salmonella Typhimurium*.³*Klebsiella* Veto Vax (Vetoquinol, Lavaltrie, QC, Canada). Active ingredients: *Klebsiella pneumoniae* bacterial extract and an emulsified adjuvant, formaldehyde, and polymyxin-B as preservatives.⁴Novodry (Zoetis Inc., Parsippany, NJ). Active ingredients: novobiocin sodium and procaine penicillin G monohydrate.⁵Spectramast DC (Zoetis Inc., Parsippany, NJ). Active ingredient: ceftiofur hydrochloride.

60 to 380 lactating cows per farm, with 1 to 8 AMS units per farm. The mean quarter SCS was 2.5 (SD = 2.3). Mean cow-level milk production was 39.4 kg/d (SD = 10.9 kg/d). The mean quarter-level fat, protein, and lactose in milk per herd are described in Table 1. Cows were housed in conventional freestall barns without pasture access. Four of the 5 farms used iodine solution for post-milking teat disinfection; the fifth farm applied a lactic acid-based foam product.

Data Characteristics

The dataset structure is fully described in Supplemental Table S4 (see Notes). The sampling period lasted 14 mo, from December 2018 to March 2020. In total, 21,467 quarter milk samples from 1,537 mammary gland quarters of 387 cows were collected. The cows recruited were mainly multiparous cows (n = 307), with a lower number (n = 80) of primiparous cows. Cows were sampled between 3 and 305 DIM, with a mean (SD) of 118 (76) DIM. The mean (SD) period between samplings was 15 d (3 d); it was exactly 14 d for 15,985 (80.2%) of the samples collected.

Among these milk samples, 12,099 (56.4%) yielded no microorganism growth, 8,269 (38.6%) yielded a growth of 1 (n = 6,901; 32.2%) or 2 (n = 1,368; 6.4%) phenotypically different colony-forming units, 1,068 (5.0%) were considered contaminated (uninformative), and 26 (0.1%) samples had missing bacteriological culture results.

Among samples where IMI status could be determined (i.e., after excluding contaminated or missing samples), there were 25 different staphylococcal species retrieved (the full list is available in Supplemental Table S2 [see Notes]). The most frequently isolated species were *S. chromogenes* (n = 835; 4.1% of interpretable samples), *S. epidermidis* (n = 530; 2.6% of interpretable samples), *S. haemolyticus* (n = 423; 2.1% of interpretable samples), *S.*

aureus (n = 389; 1.9% of interpretable samples), *S. simulans* (n = 349; 1.7% of interpretable samples), *S. xylosus* (n = 172; 0.8% of interpretable samples), *S. equorum* (n = 138; 0.7% of interpretable samples), *S. auricularis* (n = 128; 0.6% of interpretable samples), and *S. hominis* (n = 113; 0.6% of interpretable samples).

Estimating the Apparent Prevalence of Staphylococcal Species IMI During the Lactating Period

The estimated IMI apparent prevalence during lactation for the most frequent staphylococcal IMI are reported in Table 2. Among all cows (primiparous, n = 80 and multiparous, n = 307), the most prevalent staphylococcal IMI were *S. chromogenes*. (4.1%, 95% CI: 3.8%–4.4%), *S. epidermidis* (2.6%, 95% CI: 2.4%–2.8%), *S. haemolyticus* (2.1%, 95% CI: 1.9%–2.3%) and *S. aureus* (1.9%, 95% CI: 1.8%–2.1%).

The DIM-specific prevalence estimations are presented in Figure 1 for the different staphylococcal species. For most species, the IMI apparent prevalence was relatively stable throughout the lactation cycle, with some species showing minor increases (e.g., *S. aureus*, *S. auricularis*, *S. epidermidis*, *S. haemolyticus*, *S. simulans*) or decreases (e.g., *S. chromogenes*, *S. equorum*) in prevalence as DIM increased.

When investigating staphylococcal species prevalence as function of DIM and parity, only the *S. xylosus* model yielded significant parity × DIM interaction terms, indicating that the apparent prevalence of these IMI varies between primiparous and multiparous cows, but also as function of DIM, and that this latter variation was not the same in primiparous versus multiparous cows. The prevalence of *S. xylosus* IMI was generally higher in multiparous cows. For instance, an apparent prevalence of 0.2% (95% CI: 0.1%–0.6%) was estimated in primipa-

Table 2. Mean estimates of IMI apparent prevalence, incidence and elimination rates during the lactation for the most frequent staphylococci species, in quarters of cows from 5 farms using an AMS (Québec, Canada)¹

Species	Apparent prevalence, % (95% CI)	Incidence rate, NIMI/100 quarters-week (95% CI)	Elimination rate, EIMI/100 quarters-week (95% CI)
<i>Staphylococcus aureus</i>	1.9 (1.8–2.1)	0.2 (0.2–0.3)	9.6 (7.7–11.9)
<i>Staphylococcus auricularis</i>	0.6 (0.5–0.8)	0.3 (0.3–0.4)	49.7 (47.7–50.0)*
<i>Staphylococcus chromogenes</i>	4.1 (3.8–4.4)	0.3 (0.2–0.3)	8.4 (7.1–9.9)
<i>Staphylococcus epidermidis</i>	2.6 (2.4–2.8)	0.5 (0.4–0.6)	18.9 (16.6–21.2)
<i>Staphylococcus equorum</i>	0.7 (0.6–0.8)	0.3 (0.2–0.3)	49.1 (46.6–49.8)*
<i>Staphylococcus haemolyticus</i>	2.1 (1.9–2.3)	0.8 (0.7–0.9)	37.7 (35.3–39.8)
<i>Staphylococcus hominis</i>	0.6 (0.5–0.7)	0.3 (0.2–0.3)	48.1 (45.2–49.3)*
<i>Staphylococcus simulans</i>	1.7 (1.5–1.9)	0.1 (0.1–0.1)	5.3 (3.7–7.3)
<i>Staphylococcus xylosus</i>	0.8 (0.7–1.0)	0.1 (0.1–0.1)	13.4 (10.0–17.3)

¹NIMI: new intramammary infections; EIMI: eliminated intramammary infections.

*Almost all observed IMI for these species were eliminated over the typical 14-d follow-up period (118 of 119 EIMI for *S. auricularis*, 117 of 119 EIMI for *S. equorum*, and 91 of 95 EIMI for *S. hominis*).

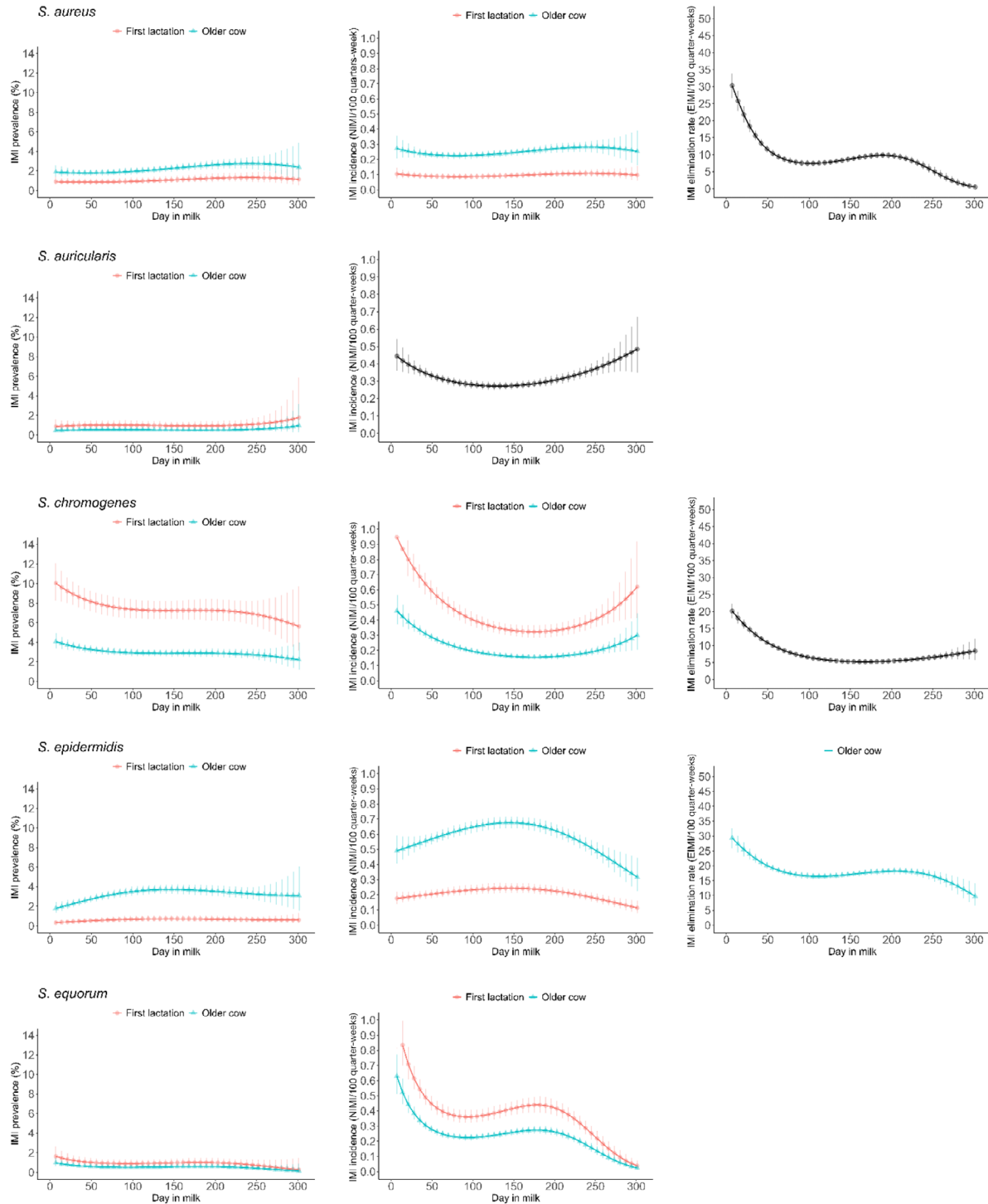


Figure 1. Quarter-level apparent prevalence (left), incidence (middle), and elimination rates (right) of staphylococci IMI as a function of DIM and parity in quarters from primiparous (n = 80) and multiparous (n = 307) dairy cows from 5 farms using AMS (Québec, Canada). Epidemiological measures are plotted by parity when relevant. Error bars represent the 95% CI. To facilitate visualization and comparison between species, prevalence plots were limited to values between 0% and 14%, incidence rate plots were limited to values between 0 and 1 new IMI/100 quarter-weeks, and elimination rate plots were limited to values between 0 and 50 eliminated IMI/100 quarter-weeks (as a result, a few 95% CI that were exceeding these values are not fully presented on the figures). Elimination rates plots are not available for *Staphylococcus auricularis*, *Staphylococcus equorum*, and *Staphylococcus hominis* because almost all observed IMI for these species were eliminated over the typical 14 d follow-up period (117 of 119 eliminated IMI for *S. equorum*, 118 of 119 eliminated IMI for *S. auricularis*, and 91 of 95 eliminated IMI for *S. hominis*). Similarly, the *Staphylococcus epidermidis* IMI elimination rate in primiparous cows is not illustrated because only 16 IMIs were available for this group. NIMI = new intramammary infections; EIMI = eliminated intramammary infections.

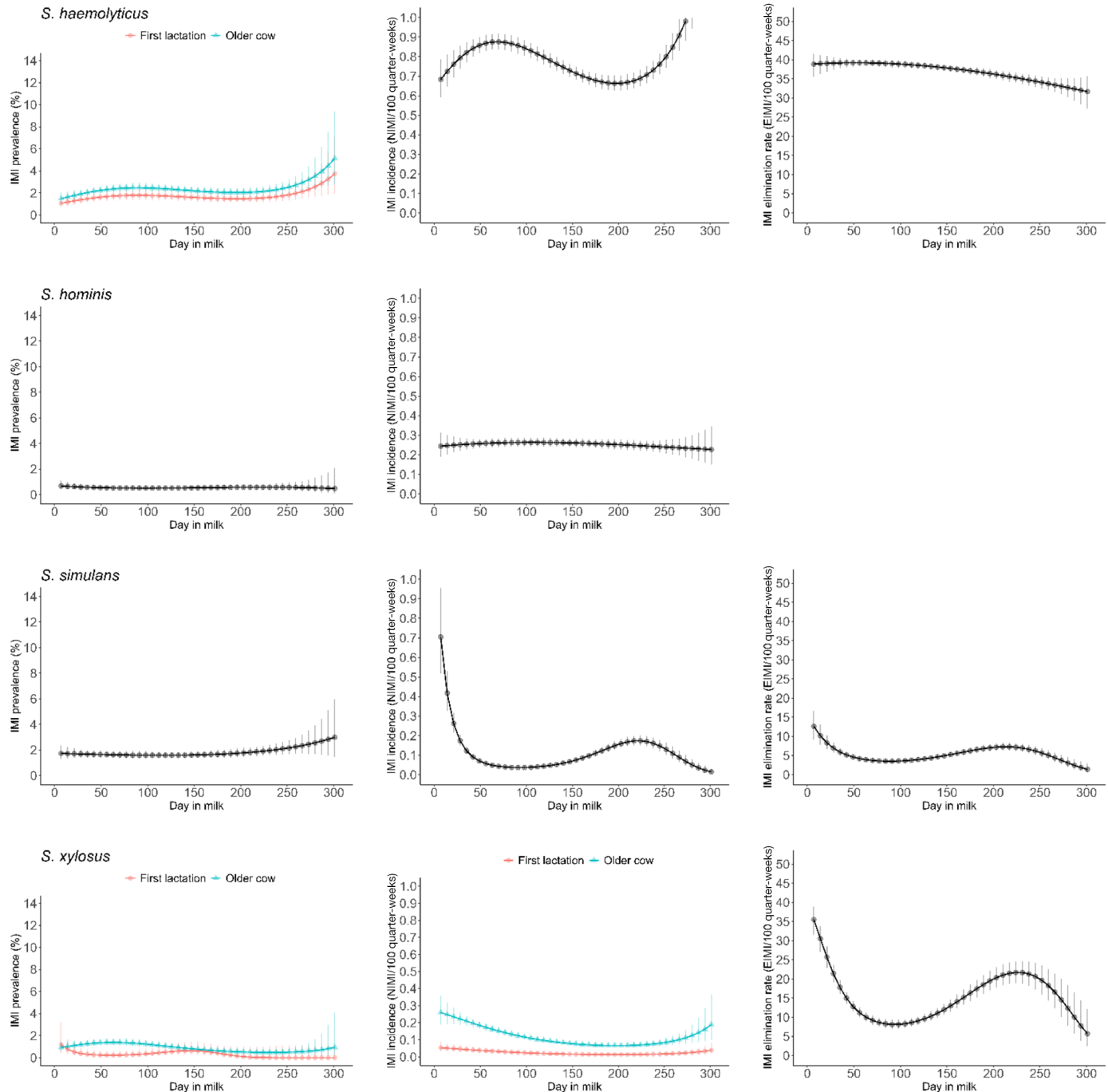


Figure 1 (Continued). Quarter-level apparent prevalence (left), incidence (middle), and elimination rates (right) of staphylococci IMI as a function of DIM and parity in quarters from primiparous ($n = 80$) and multiparous ($n = 307$) dairy cows from 5 farms using AMS (Québec, Canada). Epidemiological measures are plotted by parity when relevant. Error bars represent the 95% CI. To facilitate visualization and comparison between species, prevalence plots were limited to values between 0% and 14%, incidence rate plots were limited to values between 0 and 1 new IMI/100 quarter-weeks, and elimination rate plots were limited to values between 0 and 50 eliminated IMI/100 quarter-weeks (as a result, a few 95% CI that were exceeding these values are not fully presented on the figures). Elimination rates plots are not available for *Staphylococcus auricularis*, *Staphylococcus equorum*, and *Staphylococcus hominis* because almost all observed IMI for these species were eliminated over the typical 14 d follow-up period (117 of 119 eliminated IMI for *S. equorum*, 118 of 119 eliminated IMI for *S. auricularis*, and 91 of 95 eliminated IMI for *S. hominis*). Similarly, the *Staphylococcus epidermidis* IMI elimination rate in primiparous cows is not illustrated because only 16 IMIs were available for this group. NIMI = new intramammary infections; EIMI = eliminated intramammary infections.

rous cows at 70 DIM, whereas a prevalence of 1.4% (95% CI: 1.1%–1.7%) was estimated for that same production cycle period in multiparous cows.

For the models describing the apparent prevalence of IMI by *S. aureus*, *S. auricularis*, *S. chromogenes*, *S. epidermidis*, *S. equorum*, and *S. haemolyticus*, the parity $\times$ DIM interaction terms were not significant and were therefore removed, but the parity main term was significant, indicating that the prevalence of these IMI varied as a function of the cows' age. Intramammary infections by *S. aureus*, *S. epidermidis*, and *S. haemolyticus* were more frequent in multiparous cows. For instance, at 70 DIM, we estimated *S. aureus* IMI apparent prevalence of 0.9% (95% CI: 0.6%–1.2%) in primiparous and 1.8% (95% CI: 1.6%–2.2%) in multiparous cows; *S. epidermidis* IMI apparent prevalence of 0.6% (95% CI: 0.4%–0.9%) in primiparous and 3.1% (95% CI: 2.7%–3.6%) in multiparous cows; and *S. haemolyticus* IMI apparent prevalence of 1.8% (95% CI: 1.3%–2.3%) in primiparous and 2.4% (95% CI: 2.1%–2.8%) in multiparous cows.

On the other hand, IMI by *S. auricularis*, *S. chromogenes*, and *S. equorum* were more frequent in primiparous cows. For instance, at 70 DIM, we estimated *S. auricularis* IMI apparent prevalence of 1.0% (95% CI: 0.7%–1.5%) in primiparous and 0.6% (95% CI: 0.4%–0.7%) in multiparous cows; *S. chromogenes* IMI apparent prevalence of 7.7% (95% CI: 6.7%–8.8%) in primiparous and 3.1% (95% CI: 2.7%–3.5%) in multiparous cows; and *S. equorum* IMI apparent prevalence of 0.9% (95% CI: 0.6%–1.3%) in primiparous and 0.5% (95% CI: 0.4%–0.7%) in multiparous cows.

For the models describing the prevalence of IMI by *S. hominis* and *S. simulans*, the parity $\times$ DIM interaction, as well as the parity terms, were not significant, indicating that IMI apparent prevalence of these species was relatively similar in primiparous and multiparous cows.

Estimating Acquisition and Elimination Rates of Staphylococcal Species IMI During the Lactation

The overall estimated incidence and elimination rates for the most frequent staphylococcal IMI are presented in Table 2. Among the studied staphylococci, *S. haemolyticus* (0.8 new IMI [NIMI]/100 quarter-weeks, 95% CI: 0.7–0.9) and *S. epidermidis* (0.5 NIMI/100 quarter-weeks, 95% CI: 0.4–0.6) had the highest incidence rates. The species with the highest elimination rates were *S. auricularis* (49.7 eliminated IMI [EIMI]/100 quarter-weeks, 95% CI: 47.7–50.0), *S. equorum* (49.1 EIMI/100 quarter-weeks, 95% CI: 46.6–49.8) and *S. hominis* (48.1 EIMI/100 quarter-weeks, 95% CI: 45.2–49.3). *Staphylococcus chromogenes* (8.4 EIMI/100 quarter-weeks, 95% CI: 7.1–9.9) and *S. simulans* (5.3 EIMI/100 quarter-weeks, 95% CI: 3.7–7.3) yielded elimination rates nu-

merically similar or lower to the one of *S. aureus* (9.6 EIMI/100 quarter-weeks, 95% CI: 7.7–11.9).

The DIM-specific incidence and elimination rate estimations are presented in Figure 1. For most staphylococcal species, the estimated incidence and elimination rates varied widely as a function of DIM and, often, parity. For instance, for *S. chromogenes*, *S. equorum*, *S. simulans*, and *S. xylosus*, higher IMI incidence rates were observed in early lactation (<50 DIM) as compared with the rest of the production cycle. The IMI elimination rates of *S. aureus*, *S. chromogenes*, *S. epidermidis*, *S. simulans*, and *S. xylosus* were also higher during the <50 DIM period, as compared with later in the lactation.

The parity $\times$ DIM interaction terms were only significant for the *S. xylosus* IMI incidence and the *S. epidermidis* IMI elimination models, indicating very distinct incidence (for *S. xylosus*) or elimination rate (for *S. epidermidis*) values in primiparous and multiparous cows during their production cycle. However, the *S. epidermidis* IMI elimination rate in primiparous cows was estimated using only 16 IMI.

Parity was also a significant predictor for the models estimating the incidence rate for IMI by *S. aureus*, *S. chromogenes*, *S. epidermidis*, and *S. equorum*, as well as for the models estimating IMI elimination rates of *S. equorum* and *S. hominis*. Higher IMI incidence rates were observed in multiparous cows, as compared with primiparous cows, for *S. aureus*, *S. epidermidis*, and *S. xylosus*. On the other hand, higher IMI incidence rates were observed in primiparous cows for *S. chromogenes* and *S. equorum*.

Incidence rates were not significantly affected by parity for *S. auricularis*, *S. haemolyticus*, *S. hominis*, and *S. simulans*. Similarly, the elimination rates of IMI by *S. aureus*, *S. chromogenes*, *S. haemolyticus*, *S. simulans*, and *S. xylosus* were not associated with parity.

Estimating the Association of Staphylococcal IMI with Changes in Quarter Milk SCS, Quarter Milk Production, and Quarter Milk Components (Fat, Protein, Lactose)

We were able to describe the effect on SCS, milk yield, and milk components (fat, protein, lactose) of lactational IMI caused by *S. aureus*, *S. chromogenes*, *S. epidermidis*, *S. haemolyticus*, *S. simulans*, and *S. xylosus*. The dataset used to investigate the effect of lactational IMI by these species on quarter-level SCS during lactation consisted of 7,703 test-day samples ($n = 1,884$ collected from primiparous and $n = 5,819$ from multiparous cows), from 1,349 quarters of 368 cows (Supplemental Table S5, see Notes). For these analyses, test-day samples were collected between 3 and 305 DIM, with a mean (SD) of 117 d (72 d). In total, there were 6,117 (79.5%) quarters

categorized as healthy on the sampling day and 1,027 (13.3%) quarters categorized as having an IMI by one of the studied staphylococci in pure culture on sampling day (*S. aureus*, $n = 162$; *S. chromogenes*, $n = 361$; *S. epidermidis*, $n = 203$; *S. haemolyticus*, $n = 48$; *S. simulans*, $n = 176$; and *S. xylosus*, $n = 77$).

The mean quarter milk SCS among sampled quarters was 2.4 (SD = 2.3). The LSM of quarter milk SCS overall estimates associated with IMI caused by the most frequent staphylococcal species are presented in Table 3 and illustrated in Figure 2 (for primiparous cows) and Figure 3 (for multiparous cows). Parity, IMI status, and their interaction term were significant predictors of the quarter milk SCS ($P < 0.05$). The highest SCS in primiparous cows was observed for IMI by *S. epidermidis* (6.2, 95% CI: 3.5–8.8), *S. aureus* (5.3, 95% CI: 3.2–7.3), *S. simulans* (4.6, 95% CI: 3.1–6.2), and *S. chromogenes* (4.2, 95% CI: 3.4–4.9). All these staphylococcal species yielded SCS significantly higher than those of healthy quarters (1.5, 95% CI: 1.1–1.9). In multiparous cows, as compared with healthy quarters (2.0, 95% CI: 1.8–2.2), significantly higher SCS were observed for quarters with a positive IMI status to *S. aureus*, *S. chromogenes*, *S. epidermidis*, *S. haemolyticus*, *S. simulans*, or *S. xylosus*. The highest SCS were observed for IMI caused by *S. aureus* (7.4, 95% CI: 6.7–8.1), *S. epidermidis* (5.3, 95% CI: 4.6–5.9) and *S. simulans* (5.1, 95% CI: 4.4–5.8).

The dataset used to investigate the association of having lactational IMI by the studied species with changes in quarter-level milk production during lactation consisted of 9,967 test-day samples collected from primiparous ($n = 2,437$ test-days) and multiparous cows ($n = 7,530$ test-days), from 1,381 quarters of 368 cows (Supplemental Table S6, see Notes). Test-day samples were collected between 3 to 305 DIM, with a mean (SD) of 115 d (72 d) DIM. In total, there were 7,879 (79.0%) test-days where a quarter was classified as healthy, and 1,326 (13.3%)

test-days where a quarter was considered positive for a staphylococcal IMI (*S. aureus*, $n = 208$; *S. chromogenes*, $n = 488$; *S. epidermidis*, $n = 240$; *S. haemolyticus*, $n = 60$; *S. simulans*, $n = 236$; and *S. xylosus*, $n = 94$).

Mean quarter milk production was 9.7 kg/quarter-day (SD = 3.4). The LSM estimates of the effect of being infected with a given staphylococcal species or other IMI on quarter milk production are presented in Table 3 and illustrated in Figure 2 (for primiparous cows) and Figure 3 (for multiparous cows). Days in milk, IMI status, and parity were all statistically associated with quarter milk yield ($P < 0.05$), but the interaction terms between IMI and parity ($P = 0.83$) were not significantly associated with quarter milk yield. In primiparous cows, the lowest mean quarter milk production was observed for quarters infected with *S. aureus* (6.9 kg/d, 95% CI: 3.6–10.2 kg/d), compared, for instance, with a production of 7.8 kg/d (95% CI: 6.8–8.9 kg/d) for healthy quarters. Nevertheless, after adjusting for multiple comparisons, milk yields of the quarters from primiparous cows with different IMI statuses were not significantly different from one another. In multiparous cows, the lowest mean quarter milk production was observed for quarters infected with *S. aureus* (7.9 kg/d, 95% CI: 6.4–9.3 kg/d), whereas a milk yield of 10.2 kg/d (95% CI: 9.3–11.1 kg/d) was observed in healthy quarters (Table 3). Among multiparous cows, the milk yield of quarters infected by *S. aureus* was statistically lower than that of healthy quarters and also from the milk yield of quarters infected with *S. chromogenes*, *S. epidermidis*, *S. haemolyticus*, *S. simulans*, *S. xylosus*, or other IMI ($P < 0.05$).

The datasets used to investigate the association of having lactational IMI by the studied species with changes in quarter-level milk fat, protein, and lactose component during lactation were fairly similar and are presented in Supplemental Tables S7, S8, and S9 (see Notes). For instance, the milk fat dataset consisted of 6,159 test-day

Table 3. Least squares means estimates of the effect of being infected with a given bacterial species or group on the quarter milk SCS and quarter milk production; estimates were obtained from quarter milk sample results of cows from 5 farms using an AMS (Québec, Canada)

Species	Quarter milk mean SCS, score (95% CI)		Quarter milk mean production, kg/d (95% CI)	
	Primiparous	Multiparous	Primiparous	Multiparous
<i>Staphylococcus aureus</i>	5.3 (3.2–7.3) ^{bc}	7.4 (6.7–8.1) ^c	6.9 (3.6–10.2) ^a	7.9 (6.4–9.3) ^a
<i>Staphylococcus chromogenes</i>	4.2 (3.4–4.9) ^{bc}	4.3 (3.7–4.9) ^{bc}	7.3 (5.7–8.9) ^a	10.1 (8.7–11.4) ^b
<i>Staphylococcus epidermidis</i>	6.2 (3.5–8.8) ^c	5.3 (4.6–5.9) ^d	7.0 (3.1–10.9) ^a	9.8 (8.5–11.1) ^b
<i>Staphylococcus haemolyticus</i>	2.4 (0.6–4.2) ^{ab}	4.7 (3.6–5.7) ^{bcd}	7.4 (4.8–10.0) ^a	10.6 (8.9–12.3) ^b
<i>Staphylococcus simulans</i>	4.6 (3.1–6.2) ^{bc}	5.1 (4.4–5.8) ^{cd}	7.5 (4.1–10.8) ^a	10.8 (9.2–12.4) ^b
<i>Staphylococcus xylosus</i>	4.1 (1.3–7.0) ^{abc}	4.5 (3.6–5.4) ^{bcd}	10.2 (5.4–15.0) ^a	11.7 (10.0–13.5) ^b
Healthy	1.5 (1.1–1.9) ^a	2.0 (1.8–2.2) ^a	7.8 (6.8–8.9) ^a	10.2 (9.3–11.1) ^b
Other IMI ¹	3.2 (2.2–4.2) ^{bc}	3.9 (3.6–4.2) ^b	7.7 (6.2–9.2) ^a	10.2 (9.2–11.2) ^b

^{a–c}Within a parameter and age group, categories with different superscripts were significantly different ($P < 0.05$) after adjustment for multiple comparisons using the Tukey method.

¹All the other staphylococci and the nonstaphylococci microorganisms identified by Kurban et al. (2022) as potential mastitis pathogens.

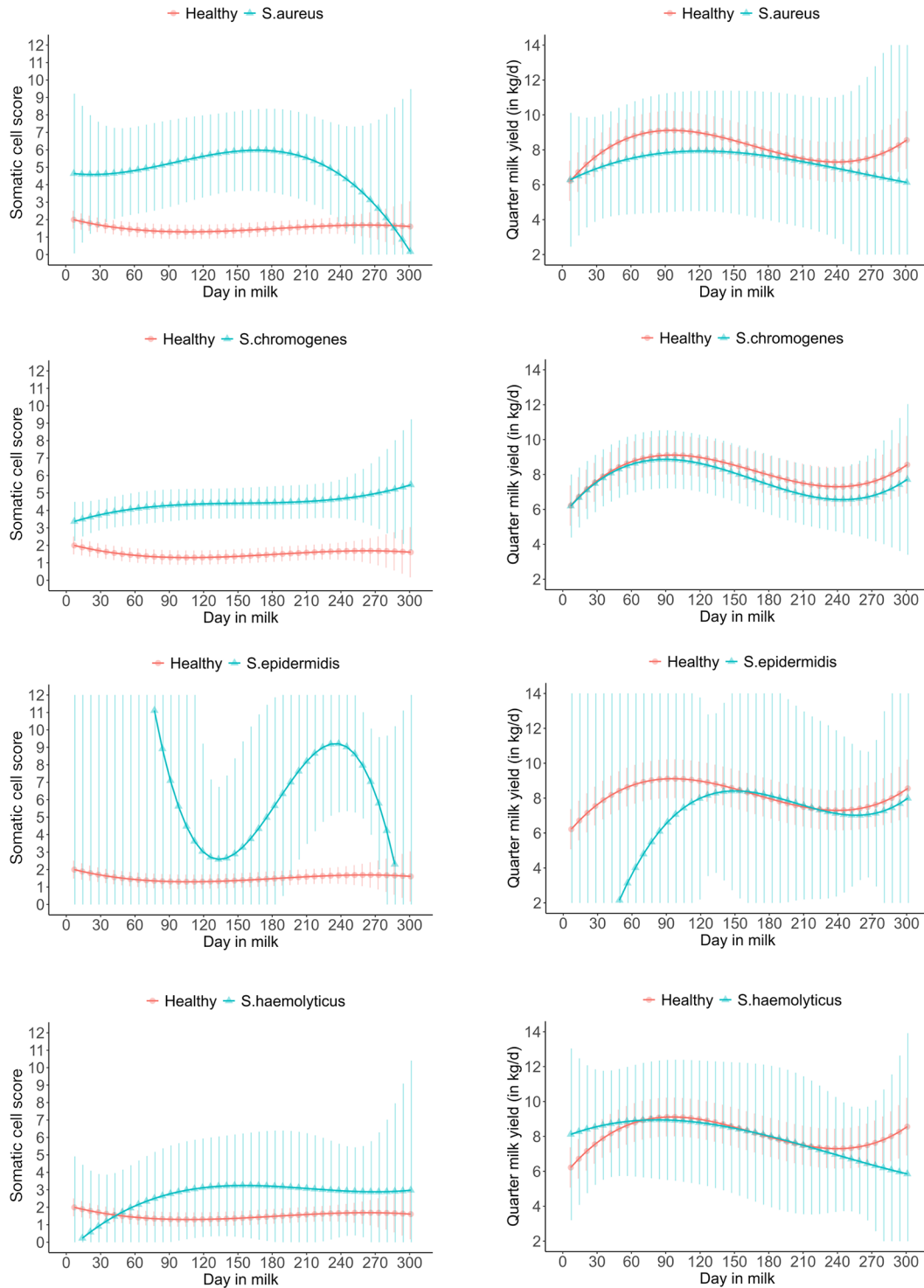


Figure 2. Effect of IMI status on lactational milk SCS (left) and quarter daily milk production (in kg/day; right) in quarters of primiparous cows ($n = 67$) classified as healthy versus infected with various *Staphylococcus* species. The LSM estimates were obtained using linear mixed models. Error bars represent the 95% CI. To facilitate visualization and comparison between species, SCS plots were limited to values between 0 and 12 and milk production plots were limited to values between 2 and 14 kg/quarter-day. As a result, a few 95% CI that were exceeding these values are not fully presented in the figures.

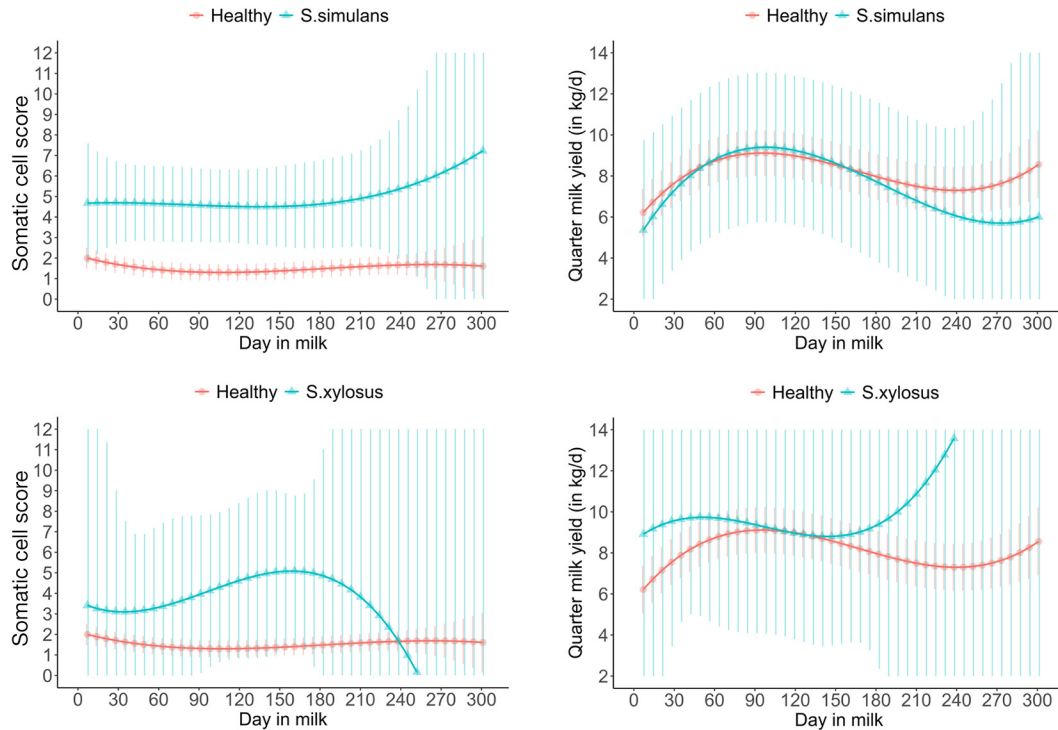


Figure 2 (Continued). Effect of IMI status on lactational milk SCS (left) and quarter daily milk production (in kg/day; right) in quarters of primiparous cows (n = 67) classified as healthy versus infected with various *Staphylococcus* species. The LSM estimates were obtained using linear mixed models. Error bars represent the 95% CI. To facilitate visualization and comparison between species, SCS plots were limited to values between 0 and 12 and milk production plots were limited to values between 2 and 14 kg/quarter-day. As a result, a few 95% CI that were exceeding these values are not fully presented in the figures.

samples collected from primiparous (n = 1,547 test-days) and multiparous cows (n = 4,612 test-days), from 1,318 quarters of 366 cows (Supplemental Table S7). Test-day samples were collected between 3 to 305 DIM, with a mean (SD) of 120 d (72 d) DIM. In total, there were 4,882 (79.2%) test-days where the quarter was classified as healthy, and 840 (13.6%) test-days where the quarter was considered positive for a staphylococcal IMI (*S. aureus*, n = 136; *S. chromogenes*, n = 293; *S. epidermidis*, n = 162; *S. haemolyticus*, n = 42; *S. simulans*, n = 151; and *S. xylosus*, n = 56).

The mean quarter milk fat content was 4.9% (SD = 2.3). The LSM estimates of the association of an IMI by given staphylococcal species or other IMI with changes in quarter milk fat content are presented in Table 4 (figures are available in the supplemental material, see Notes); DIM, IMI status and parity were all statistically associated with quarter milk fat content ($P < 0.05$). Among primiparous cows, the lowest mean quarter milk fat content was observed for quarters infected with *S. aureus* (3.9 kg/d, 95% CI: 2.3–5.5 kg/d), compared, with a milk fat content of 4.5 kg/d (95% CI: 4.0–5.0 kg/d) for healthy quarters. However, after adjusting for multiple comparisons, milk fat content of the quarters

from primiparous cows with different IMI statuses were not significantly different from one another. In multiparous cows, the lowest mean quarter milk fat content was observed for quarters infected with *S. aureus* (4.4 kg/d, 95% CI: 3.7–5.1 kg/d), whereas a milk fat content of 5.0 kg/d (95% CI: 4.7–5.4 kg/d) was observed in healthy quarters (Table 4). In multiparous cows, the milk fat of quarters infected by *S. aureus*, *S. chromogenes*, *S. epidermidis*, *S. haemolyticus*, *S. simulans*, *S. xylosus* was not statistically different than that of healthy quarters or other IMI.

The mean quarter milk protein among sampled quarters was 3.2% (SD = 0.3). The LSM quarter milk protein content overall estimates for the most frequent staphylococcal species IMI are presented in Table 4 (figures are available in the supplemental material, see Notes). Only IMI status was a significant predictor of the quarter milk protein content ($P < 0.05$). Although the lowest quarter milk protein content in primiparous cows was observed for IMI by *S. simulans* (3.1%, 95% CI: 2.8%–3.3%) and *S. xylosus* (3.1%, 95% CI: 2.6%–3.5%), no statistically significant difference was observed for quarters having any staphylococcal IMI or other IMI and healthy quarters (Table 4). In multiparous cows, unexpectedly, the highest

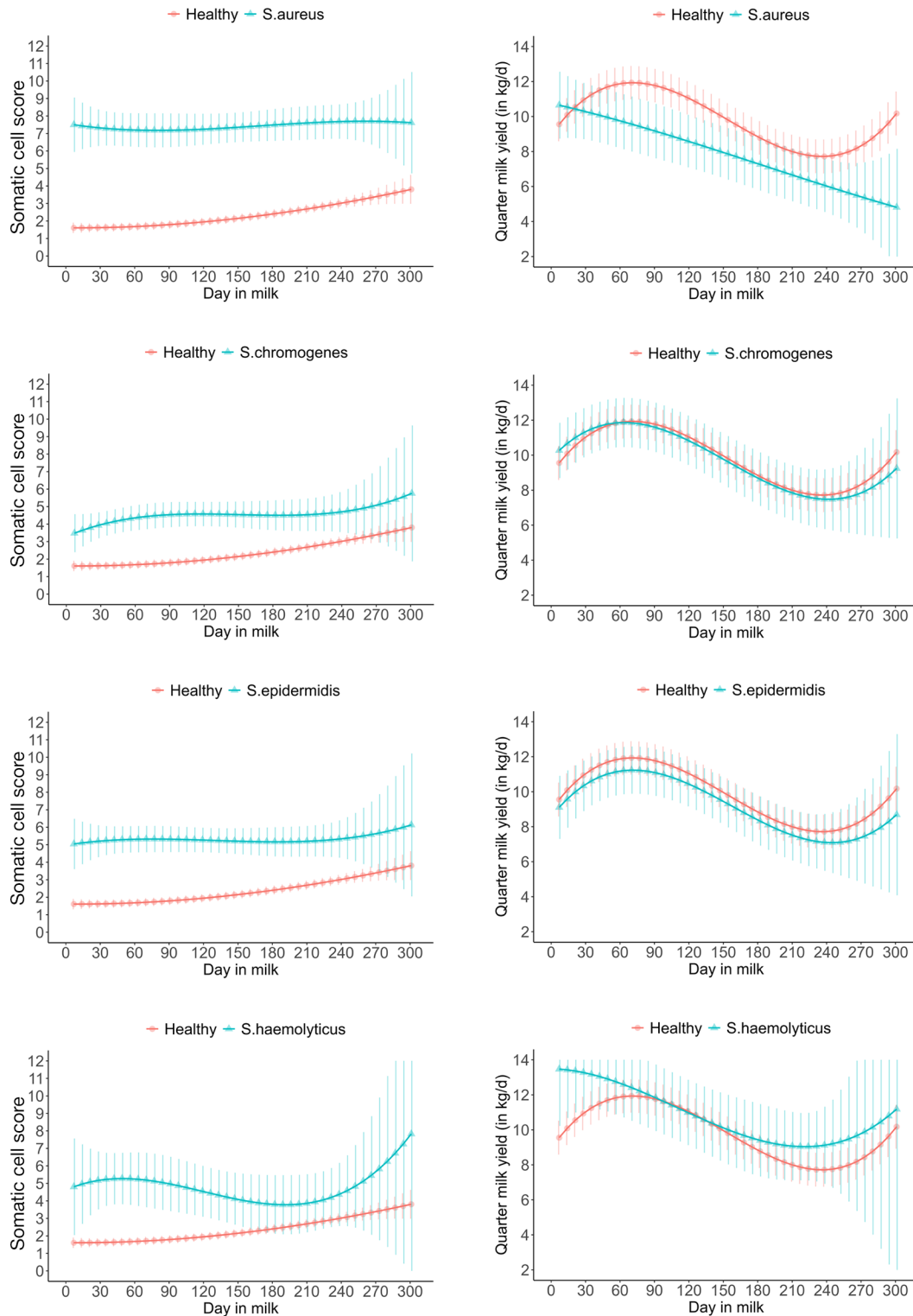


Figure 3. Effect of IMI status on lactational milk SCS (left) and quarter daily milk production (in kg/day; right) in quarters of multiparous cows ($n = 301$) classified as healthy versus infected with various *Staphylococcus* species. The LSM estimates were obtained using linear mixed models. Error bars represent the 95% CI. To facilitate visualization and comparison between species, SCS plots were limited to values between 0 and 12 and milk production plots were limited to values between 2 and 14 kg/quarter-day. As a result, a few 95% CI that were exceeding these values are not fully presented in the figures.

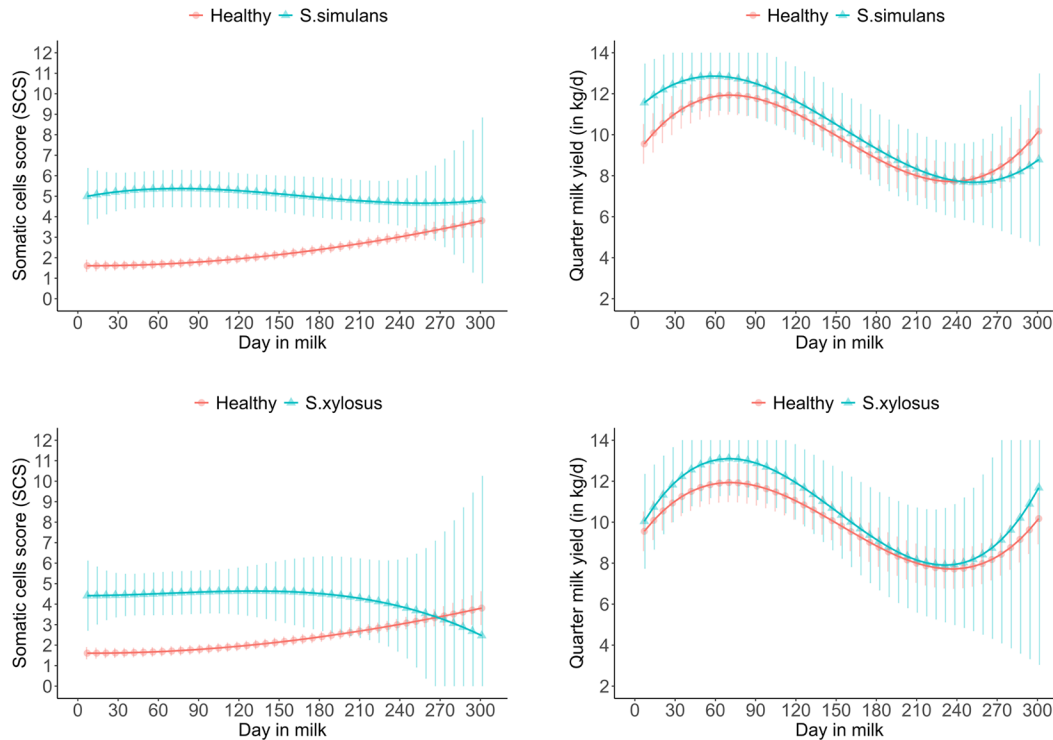


Figure 3 (Continued). Effect of IMI status on lactational milk SCS (left) and quarter daily milk production (in kg/day; right) in quarters of multiparous cows ($n = 301$) classified as healthy versus infected with various *Staphylococcus* species. The LSM estimates were obtained using linear mixed models. Error bars represent the 95% CI. To facilitate visualization and comparison between species, SCS plots were limited to values between 0 and 12 and milk production plots were limited to values between 2 and 14 kg/quarter-day. As a result, a few 95% CI that were exceeding these values are not fully presented in the figures.

quarter milk protein content was observed for IMI caused by *S. aureus* (3.4%, 95% CI: 3.2%–3.5%).

The mean quarter milk lactose among sampled quarters was 4.4% (SD = 0.3). The LSM quarter milk lactose content overall estimates for the most frequent

staphylococcal species are presented in Table 4 (figures are available in the supplemental material, see Notes). Parity, IMI status, and their interaction term were significant predictors of the quarter milk lactose content ($P < 0.05$). In primiparous cows, as compared with healthy

Table 4. Least squares means estimates of the effect of being infected with a given bacterial species or group on the quarter milk fat, protein, and lactose. Estimates were obtained from quarter milk sample results of cows from 5 farms using an AMS (Québec, Canada)

Species	Quarter milk mean fat component, % (95% CI)		Quarter milk mean protein component, % (95% CI)		Quarter milk mean lactose component, % (95% CI)	
	Primiparous	Multiparous	Primiparous	Multiparous	Primiparous	Multiparous
<i>Staphylococcus aureus</i>	3.9 (2.3–5.5) ^a	4.4 (3.7–5.1) ^{ab}	3.2 (2.9–3.5) ^a	3.4 (3.2–3.5) ^b	4.5 (4.3–4.7) ^{ab}	4.2 (4.1–4.3) ^a
<i>Staphylococcus chromogenes</i>	4.4 (3.7–5.1) ^a	4.7 (4.0–5.3) ^{ab}	3.2 (3.0–3.3) ^a	3.1 (3.0–3.3) ^a	4.5 (4.4–4.6) ^a	4.4 (4.3–4.5) ^{bc}
<i>Staphylococcus epidermidis</i>	4.2 (1.0–7.4) ^a	4.9 (4.2–5.5) ^{ab}	3.4 (3.0–3.9) ^a	3.2 (3.1–3.3) ^{ab}	4.5 (4.2–4.9) ^{ab}	4.3 (4.2–4.4) ^{ab}
<i>Staphylococcus haemolyticus</i>	4.9 (3.0–6.7) ^a	5.1 (3.9–6.3) ^{ab}	3.3 (3.0–3.5) ^a	3.2 (3.0–3.4) ^{ab}	4.6 (4.4–4.8) ^{ab}	4.3 (4.2–4.5) ^{abc}
<i>Staphylococcus simulans</i>	4.5 (3.2–5.9) ^a	4.9 (4.2–5.6) ^{ab}	3.1 (2.8–3.3) ^a	3.2 (3.0–3.3) ^a	4.4 (4.2–4.5) ^a	4.3 (4.2–4.4) ^{ab}
<i>Staphylococcus xylosus</i>	4.7 (2.2–7.2) ^a	5.2 (4.2–6.1) ^{ab}	3.1 (2.6–3.5) ^a	3.1 (2.9–3.3) ^a	4.5 (4.2–4.7) ^{ab}	4.3 (4.2–4.5) ^{abc}
Healthy	4.5 (4.0–5.0) ^a	5.0 (4.7–5.4) ^b	3.2 (3.1–3.3) ^a	3.2 (3.1–3.3) ^a	4.6 (4.5–4.7) ^b	4.4 (4.3–4.5) ^c
Other IMI ¹	3.8 (2.5–5.1) ^a	4.6 (4.1–5.0) ^a	3.2 (3.0–3.4) ^a	3.2 (3.1–3.3) ^a	4.6 (4.4–4.7) ^{ab}	4.4 (4.3–4.4) ^b

^{a–c}Within a parameter and age group, categories with different superscripts were significantly different ($P < 0.05$) after adjustment for multiple comparisons using the Tukey method.

¹All the other staphylococci and the nonstaphylococci microorganisms identified by Kurban et al. (2022) as potential mastitis pathogens.

quarters (4.6%, 95% CI: 4.5%–4.7%), slightly lower, but still statistically different, lactose content was observed for quarters with a positive IMI status to *S. chromogenes* (4.5%, 95% CI: 4.4%–4.6%) and *S. simulans* (4.4%, 95% CI: 4.2%–4.5%). Among multiparous cows, slightly lower ($P < 0.05$) quarter milk lactose content was reported for quarters having an IMI by *S. aureus* (4.2%, 95% CI: 4.1%–4.3%), *S. epidermidis* (4.3%, 95% CI: 4.2%–4.4%), and *S. simulans* (4.3%, 95% CI: 4.2%–4.4%), compared with healthy quarters (4.4%; 95% CI: 4.3%–4.5%).

The results from the sensitivity analysis where we repeated the SCS, milk yield, and composition analyses, but using a more liberal quarter IMI status definition based on evaluation of a single milk sample, are presented as in the supplemental material (see Notes). In general, the differences that were significant in the initial analysis were also significant in the sensitivity analysis. However, the estimated differences between healthy and infected quarters were usually smaller. For instance, in the sensitivity analysis, quarter milk yield in multiparous cows was estimated at 9.7 kg/d in healthy and 9.1 kg/d in *S. aureus*-infected quarters (vs. 10.2 and 7.8 kg/d in the original analyses).

DISCUSSION

An objective of this study was to investigate the species-specific apparent prevalence of lactational staphylococcal IMI in dairy cows. In our study, the 9 most frequent staphylococcal species isolated were, in descending order, *S. chromogenes*, *S. epidermidis*, *S. aureus*, *S. simulans*, *S. haemolyticus*, *S. xylosus*, *S. auricularis*, *S. equorum*, and *S. hominis*. These same species, except for *S. auricularis* and *S. hominis*, were confirmed to be the most prevalent staphylococci in other studies (Fry et al., 2014; Condas et al., 2017; Valckenier et al., 2021; Kurban et al., 2022). In fact, staphylococcal species-level IMI prevalence is expected to differ among dairy farms. This was the case in the current study, where the staphylococcal species distribution differed among the studied farms (data presented as in the supplemental material, see Notes). Actually, intraherd prevalence could also vary from one period to another, especially in open herds, where animals from other herds could be introduced (Gillespie et al., 2009). *Staphylococcus chromogenes* IMI tended to be more prevalent in primiparous cows, and *S. epidermidis* IMI were recovered more often in older cows, suggesting that the epidemiology of these species is perhaps related to the cow's life cycle. These findings are consistent with the results of research by Thorberg et al. (2009).

Another aim of our study was to describe the species-specific epidemiology of staphylococcal IMI in terms of the acquisition of new and spontaneous cures (i.e., IMI

resolved without any treatment) of existing IMI. For most staphylococcal species, the IMI apparent prevalence did not vary much throughout the lactation. On the other hand, some IMI appeared to be more readily acquired during specific periods of the production cycle. For instance, *S. chromogenes*, *S. equorum*, and *S. simulans* IMI incidence rates were at their highest during early lactation. Indeed, Valckenier et al. (2020) reported *S. chromogenes* as the most prevalent NAS species present at calving in first-lactation dairy cows. In our previous study, *S. equorum* and *S. simulans* were among the most frequently isolated NAS species in primiparous cows, as well as in multiparous cows, in early lactation (Kurban et al., 2025).

Regarding IMI elimination, we observed very high elimination rates for *S. auricularis*, *S. equorum*, *S. haemolyticus*, and *S. hominis*, suggesting that these species more frequently cause transient, rather than persistent, IMI. On the other hand, *S. aureus*, *S. chromogenes*, and *S. simulans* IMI appeared to be equally able to persist in the mammary gland. Fry et al. (2014) also observed shorter IMI duration for *S. epidermidis*, *S. haemolyticus*, and *S. xylosus*, and longer IMI durations for *S. simulans* and *S. chromogenes*. In our study, however, higher elimination rates were sometimes observed for IMI occurring during early lactation. For instance, we observed relatively higher early lactation elimination rates for *S. aureus*, *S. chromogenes*, *S. epidermidis*, and *S. simulans* among primiparous and multiparous cows. These findings are consistent with other studies. For instance, Skoulikas et al. (2018) observed cure rates of 34 EIMI/100 quarters-month in nontreated quarters from primiparous cows infected by *S. aureus* during the beginning of the lactation. In our research, parity was not statistically associated with *S. aureus* IMI elimination. However, most of our early lactation *S. aureus* EIMI were observed in primiparous cows (data not presented), suggesting that heifers could have been infected before their first calving with a less persistent *S. aureus* strain, which was not necessarily adapted to persist in the lactating udder.

Finally, to answer our main objective, we investigated the species-specific association of lactational staphylococcal IMI with changes in udder health indicator (i.e., SCS), milk production, and components (fat, protein, lactose), including variability associated with parity. In general, we identified that the quarter IMI status is often associated with differences in the quarter SCS but, less commonly, the quarter milk yield, fat, protein, or lactose content. Lactational IMI by *S. aureus*, *S. chromogenes*, *S. epidermidis*, and *S. simulans* caused a measurable local immune response (i.e., increase in SCS, as compared with healthy quarters in both primiparous and multiparous cows). Djabri et al. (2002), in their meta-analysis, concluded that the geometric mean SCC in quarters in-

infected with *S. aureus* was 357,000 cells/mL, and 138,000 cells/mL in quarters infected by NAS. Geometric means of SCC of 657,600 cells/mL during the infection and of 317,500 cells/mL for the whole lactation in quarters infected with persistent NAS were reported by Taponen et al. (2007). Furthermore, in a recent study, Taponen et al. (2022) observed species-specific geometric means of quarter milk SCC of 131,000 cells/mL and of 216,000 cells/mL in quarters infected by *S. chromogenes* and *S. simulans*, respectively. However, quarters with IMI caused by *S. epidermidis* yielded a geometric mean of only 80,000 cells/mL (Taponen et al., 2022), which is lower than what was observed in our study.

Despite the elevated SCS associated with some staphylococcal species, in most cases, we could not highlight a negative association with the quarter milk yield, fat, protein, or lactose content. Lower production was observed only in quarters from multiparous cows infected with *S. aureus* IMI. These quarters produced, on average, 2.3 kg/d less milk than the average healthy quarter. Only a few researchers have evaluated quarter-level milk production decreases associated with subclinical mastitis or IMI caused by *S. aureus*. Gonçalves et al. (2018) reported a substantial quarter milk yield reduction of up to 0.7 kg/milking (i.e., given that cows selected for this study were milked twice, it assumingly equals to 1.4 kg/d) in quarters with confirmed subclinical mastitis caused by *S. aureus*. However, Botaro et al. (2015) estimated only a 0.1 kg (i.e., given that cows selected for this study were milked twice, it assumingly equals to 0.2 kg/d) loss of 3.5% fat-corrected milk production per milking in infected quarters on the sampling day. In addition, 2 independent studies reported no changes in quarter milk fat or protein in association with staphylococcal IMI (Botaro et al., 2015; Tomazi et al., 2015), which coincides with our findings.

The fact that we did not observe a substantial milk loss in quarters infected with staphylococcal species other than *S. aureus* could be explained in several ways. For some staphylococcal species (e.g., *S. auricularis*, *S. equorum*, *S. haemolyticus*, *S. hominis*), we observed mostly transient IMI. In addition, their presence did not induce a major elevation in SCS of the infected quarters. Therefore, a reduction in milk yield due to these IMI would be surprising, given that these IMI would mostly be eliminated before substantially harming the mammary epithelial cells. Other staphylococcal species (e.g., *S. simulans*, *S. epidermidis*, *S. chromogenes*) appeared to be able to persist for a longer period in the mammary gland and also induced a greater elevation in SCS, at least in multiparous cows. Nevertheless, we could not highlight a reduction in milk production in quarters infected with these latter species. Alternatively, we could hypothesize that some NAS species have a lower potential for invading

mammary epithelial cells, compared with *S. aureus*, for instance, and thus may cause less damage to the milk producing cells (Taponen and Pyorala, 2009).

Similar to our findings, Tomazi et al. (2015) observed no quarter milk yield changes caused by NAS as a group or at the species level for *S. chromogenes*, *S. haemolyticus*, *S. simulans*, or others. Even though the measures provided by Tomazi et al. (2015) were from a single milking, their choice of IMI case definition was slightly more conservative than in our study (i.e., the presence of an IMI by NAS species was only considered if 2 consecutive culture-positive samples for a given NAS species with 1,000 cfu/mL was obtained; a quarter was considered healthy if 2 consecutive culture-negative for any microorganism and with SCC < 200,000 cells/mL was observed).

Conversely to the current results, in a previous study, we observed that, in addition to postcalving IMI by *S. aureus*, *S. epidermidis* IMI were associated with a subsequently sustained quarter milk yield reduction of -1.8 to -2.0 kg/quarter-day (Kurban et al., 2025). In comparison to the IMI observed during lactation, we could hypothesize that the postcalving IMI may cause substantial damage to the mammary epithelial cells during a critical period of their development. This could explain the sustained impact on the production potential of the affected quarters.

Our results highlight that lactational IMI caused by *S. aureus* and *S. chromogenes* in primiparous cows, and by *S. aureus*, *S. epidermidis*, *S. simulans* in multiparous cows could induce an important SCC elevation in affected quarters during lactation. These staphylococcal species were among the most prevalent, and some are known to be contagious. If many cows in a herd are infected by *S. aureus*, *S. chromogenes*, *S. epidermidis*, or *S. simulans* and thus have a substantial immune response, this could certainly affect the bulk tank milk SCC. Preventing spread, and possibly improving elimination of these IMI in a herd, would be important to improve bulk milk quality. Still, further analyses and research could be recommended to confirm the species-specific effect of the most common staphylococcal IMI on milk production. In the present study, even with >18,000 test-day results from nearly 1,500 quarters of 380 cows, we were unable to highlight a milk production difference between quarters infected with various staphylococcal species and healthy quarters. However, our results suggest that for many species, the milk yield reduction would be from <0.1 to 0.4 kg/quarter-day, and thus probably <1% to 6% and 1% to 4% milk yield reduction for quarters of primiparous and multiparous cows, respectively.

Although most studies report either no effect or a reduction in milk yield associated with NAS intramammary infections, a limited number of observational

analyses have suggested that cows or quarters infected with certain NAS species may exhibit similar or even numerically higher milk production compared with uninfected counterparts (Piepers et al., 2013). For example, longitudinal studies in dairy heifers have shown that intramammary infection with species such as *S. chromogenes* was not associated with decreased quarter milk yield despite elevated somatic cell counts (Valckenier et al., 2020; 2021). In some datasets, NAS-infected animals produced comparable or slightly greater amounts of milk than noninfected herd mates, although these differences were not consistently statistically significant and were not interpreted as a causal stimulatory effect of infection. Such findings may reflect confounding by cow-level factors (e.g., higher-producing animals being more susceptible to colonization), reverse causality, or misclassification of persistent colonization versus transient infection. Overall, the available literature does not provide strong evidence that NAS IMI increases milk yield per se, but rather indicates that many NAS species behave as minor pathogens whose impact on production is limited or highly context-dependent.

Our study is one of very few to investigate species-specific quarter-level lactational staphylococcal IMI apparent prevalence, incidence, and elimination rates and, most importantly, the relationship of these IMI with quarter-level SCS, milk yield, fat, protein, and lactose during the lactating period. The strengths of our study include the long follow-up period, the quarter-level measurements, species-level identification, and the relatively large number of cows followed. However, our study also has its limitations. The most important limitation is the inclusion of only 5 herds with a particular type of dairy farm management (i.e., freestall barn with AMS). As a consequence, our results might not be representative of the situation on all dairy farms in Canada or worldwide. This would especially be true for our estimates of IMI apparent prevalence, which likely vary from one farm to another. However, the staphylococcal IMI effects on quarter SCS and on quarter milk production reported would likely be applicable to other Holstein cows housed or milked using different systems.

Another limitation of the current study was that we were not able to obtain confirmation (using strain typing methods, for instance) that isolates from the same bacterial species identified in a mammary gland in ≥ 2 subsequent milk samples were indeed due to a persistent IMI (as compared with a cure followed by a new IMI caused by the same bacterial species). This was not possible due to the large number of isolates that would have had to be characterized. This limitation could have led to an underestimation of the elimination and incidence rates reported. Nevertheless, in a recent study using data assembled from the same cows, herds, and period, we were

able to include strain typing (whole-genome sequencing) of bacterial isolates from the same species retrieved within a mammary gland before and after the dry period (Kurban et al., 2025). In that latter study, we observed that a persistent IMI (i.e., the same bacterial strain) was present in 92% of quarters in which the same bacterial species was retrieved before and after the dry period. These observations cannot be directly extrapolated to the lactation period, but we could hypothesize that a similar or greater proportion of strain similarity would be observed between isolates from the same species collected 2 weeks apart during lactation (as compared with 2 mo apart, before vs. after the dry period).

CONCLUSIONS

The most prevalent staphylococcal species isolated from quarters with no clinical signs of mastitis were *S. chromogenes*, *S. epidermidis*, *S. aureus*, *S. simulans*, *S. haemolyticus*, *S. xylosus*, *S. auricularis*, *S. equorum*, and *S. hominis*. In general, the majority of staphylococcal IMI were transient, with high species-specific elimination rates. Intramammary infections caused by *S. aureus*, *S. chromogenes*, *S. simulans*, and *S. xylosus*, on the other hand, had low elimination rates, especially in the mid- and late-lactation period. In primiparous cows, only *S. aureus*, *S. chromogenes*, *S. epidermidis*, and *S. simulans* IMI led to an elevated SCS but no association was found with alterations in quarter milk yield, fat, or protein. In multiparous cows, higher SCS were observed in quarters infected by *S. chromogenes*, *S. epidermidis*, *S. aureus*, *S. simulans*, *S. haemolyticus*, and *S. xylosus*, but changes in milk yield were only observed in quarters having *S. aureus* IMI. A slightly lower and statistically different milk lactose content percentage was detected, as compared with healthy quarters, in quarters having *S. chromogenes* and *S. simulans* IMI in primiparous cows, and in quarters with *S. aureus*, *S. epidermidis*, and *S. simulans* IMI in multiparous cows.

NOTES

This research is funded by a contribution from Dairy Research Cluster 3 (Dairy Farmers of Canada, Ottawa, ON, Canada; and Agriculture and Agri-Food Canada, Ottawa, ON, Canada) under the Canadian Agricultural Partnership AgriScience Program, by the Mastitis Network (Saint-Hyacinthe, QC, Canada), and by the senior author's Natural Sciences and Engineering Research Council of Canada Discovery grant (RGPIN-2020-05237; Ottawa, ON, Canada) and Canada Foundation for Innovation John R. Evans Leaders Fund (#43753). The first author was also supported by the NSERC-CREATE program on milk quality, research groups Op+lait and

GRESABO. A special acknowledgment to the dairy producers who participated in our project. The authors would like to acknowledge the participation of Ariane France (University of Guelph, Guelph, ON, Canada), Caroline Forest (Université de Montréal, Saint-Hyacinthe, QC, Canada), Méliissa Desautels (Université de Montréal, Saint-Hyacinthe, QC, Canada), Caroline Chénard (Université de Montréal, Saint-Hyacinthe, QC, Canada) and Jean-Philippe Pelletier (Université de Montréal, Saint-Hyacinthe, QC, Canada), as well as summer students, for their help with data collection, milk sampling and bacteriological culture analysis. Thank you to the Lactanet (Sainte-Anne-de-Bellevue, QC, Canada) milk quality laboratory supervisor and technical staff for the milk SCC analyses. Supplemental material for this article is available at <https://doi.org/10.5683/SP3/HZ7S1P>. Author contributions: conceptual development of study design and methodology: all authors (DK, SD, JPR, TJDV, GPK, PRFA and JRM); data collection and sampling: DK; bacteriological and MALDI-TOF analysis: DK; data curation and formal statistical analysis: SD, DK; writing (original draft preparation): DK, with support of SD and JPR; writing (reviewing and editing): all authors. All authors have contributed to and critically reviewed the article and agreed to the submitted version of the manuscript. The study was approved by the Animal Use Ethics Committee of the Université de Montréal (protocol 18-Rech-1975). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AMS = automated milking system; CM = clinical mastitis; EIMI = eliminated IMI; NIMI = new IMI.

REFERENCES

- Aghamohammadi, M., D. Haine, D. F. Kelton, H. W. Barkema, H. Hogeveen, G. P. Keefe, and S. Dufour. 2018. Herd-level mastitis-associated costs on Canadian dairy farms. *Front. Vet. Sci.* 5:100. <https://doi.org/10.3389/fvets.2018.00100>.
- Botaro, B. G., C. S. Cortinhas, A. G. Dibbern, L. F. P. Silva, N. R. Benites, and M. V. dos Santos. 2015. *Staphylococcus aureus* intramammary infection affects milk yield and SCC of dairy cows. *Trop. Anim. Health Prod.* 47:61–66. <https://doi.org/10.1007/s11250-014-0683-5>.
- Cameron, M., H. W. Barkema, J. De Buck, S. De Vliegheer, M. Chaffer, J. Lewis, and G. P. Keefe. 2017. Identification of bovine-associated coagulase-negative staphylococci by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry using a direct transfer protocol. *J. Dairy Sci.* 100:2137–2147. <https://doi.org/10.3168/jds.2016-12020>.
- Cameron, M., J. Perry, J. R. Middleton, M. Chaffer, J. Lewis, and G. P. Keefe. 2018. Short communication: Evaluation of MALDI-TOF mass spectrometry and a custom reference spectra expanded database for the identification of bovine-associated coagulase-negative staphylococci. *J. Dairy Sci.* 101:590–595. <https://doi.org/10.3168/jds.2017-13226>.
- Chen, Y., Y. Ma, Q. Ji, X. Yang, X. Feng, R. Yao, X. Cheng, T. Li, Y. Wang, and Z. Wang. 2021. Intracellular *Staphylococcus aureus* infection decreases milk protein synthesis by preventing amino acid uptake in bovine mammary epithelial cells. *Front. Vet. Sci.* 8:756375. <https://doi.org/10.3389/fvets.2021.756375>.
- Čitek, J., M. Brzřáková, L. Hanusová, O. Hanuš, L. Večerek, E. Samková, E. Jozová, I. Hoštičková, J. Trávníček, M. Klotzda, and L. Haseňová. 2022. Somatic cell score: gene polymorphisms and other effects in Holstein and Simmental cows. *Anim. Biosci.* 35:13–21. <https://doi.org/10.5713/ab.20.0720>.
- Condas, L. A. Z., J. De Buck, D. B. Nobrega, D. A. Carson, S. Naushad, S. De Vliegheer, R. N. Zadoks, J. R. Middleton, S. Dufour, J. P. Kastelic, and H. W. Barkema. 2017. Prevalence of non-aureus staphylococci species causing intramammary infections in Canadian dairy herds. *J. Dairy Sci.* 100:5592–5612. <https://doi.org/10.3168/jds.2016-12478>.
- Coulon, J.-B., P. Gasqui, J. Barnouin, A. Ollier, P. Pradel, and D. Pomiès. 2002. Effect of mastitis and related-germ on milk yield and composition during naturally-occurring udder infections in dairy cows. *Anim. Res.* 51:383–393. <https://doi.org/10.1051/animres:2002031>.
- De Buck, J., V. Ha, S. Naushad, D. B. Nobrega, C. Luby, J. R. Middleton, S. De Vliegheer, and H. W. Barkema. 2021. Non-aureus staphylococci and bovine udder health: Current understanding and knowledge gaps. *Front. Vet. Sci.* 8:658031. <https://doi.org/10.3389/fvets.2021.658031>.
- De Visscher, A., S. Piepers, F. Haesebrouck, and S. De Vliegheer. 2016. Intramammary infection with coagulase-negative staphylococci at parturition: Species-specific prevalence, risk factors, and effect on udder health. *J. Dairy Sci.* 99:6457–6469. <https://doi.org/10.3168/jds.2015-10458>.
- Deluyker, H. A., R. H. Shumway, W. E. Wecker, A. S. Azari, and L. D. Weaver. 1990. Modeling daily milk-yield in holstein cows using time-series analysis. *J. Dairy Sci.* 73:539–548. [https://doi.org/10.3168/jds.S0022-0302\(90\)78701-2](https://doi.org/10.3168/jds.S0022-0302(90)78701-2).
- Djabri, B., N. Bareille, F. Beaudeau, and H. Seegers. 2002. Quarter milk somatic cell count in infected dairy cows: A meta-analysis. *Vet. Res.* 33:335–357. <https://doi.org/10.1051/vetres:2002021>.
- Dohoo, I., S. Andersen, R. Dingwell, K. Hand, D. Kelton, K. Leslie, Y. Schukken, and S. Godden. 2011a. Diagnosing intramammary infections: Comparison of multiple versus single quarter milk samples for the identification of intramammary infections in lactating dairy cows. *J. Dairy Sci.* 94:5515–5522. <https://doi.org/10.3168/jds.2011-4486>.
- Dohoo, I. R., J. Smith, S. Andersen, D. F. Kelton, S. Godden, and Mastitis Research Workers' Conference. 2011b. Diagnosing intramammary infections: Evaluation of definitions based on a single milk sample. *J. Dairy Sci.* 94:250–261. <https://doi.org/10.3168/jds.2010-3559>.
- Dufour, S., I. R. Dohoo, H. W. Barkema, L. Descoteaux, T. J. Devries, K. K. Reyher, J. P. Roy, and D. T. Scholl. 2012a. Epidemiology of coagulase-negative staphylococci intramammary infection in dairy cattle and the effect of bacteriological culture misclassification. *J. Dairy Sci.* 95:3110–3124. <https://doi.org/10.3168/jds.2011-5164>.
- Dufour, S., I. R. Dohoo, H. W. Barkema, L. Descoteaux, T. J. Devries, K. K. Reyher, J. P. Roy, and D. T. Scholl. 2012b. Manageable risk factors associated with the lactational incidence, elimination, and prevalence of *Staphylococcus aureus* intramammary infections in dairy cows. *J. Dairy Sci.* 95:1283–1300. <https://doi.org/10.3168/jds.2011-4711>.
- Fry, P. R., J. R. Middleton, S. Dufour, J. Perry, D. Scholl, and I. Dohoo. 2014. Association of coagulase-negative staphylococcal species, mammary quarter milk somatic cell count, and persistence of intramammary infection in dairy cattle. *J. Dairy Sci.* 97:4876–4885. <https://doi.org/10.3168/jds.2013-7657>.
- Gillespie, B. E., S. I. Headrick, S. Boonyayatra, and S. P. Oliver. 2009. Prevalence and persistence of coagulase-negative *Staphylococcus* species in three dairy research herds. *Vet. Microbiol.* 134:65–72. <https://doi.org/10.1016/j.vetmic.2008.09.007>.
- Gonçalves, J. L., C. Kamphuis, C. M. M. R. Martins, J. R. Barreiro, T. Tomazi, A. H. Gameiro, H. Hogeveen, and M. V. dos Santos. 2018. Bovine subclinical mastitis reduces milk yield and economic return. *Livest. Sci.* 210:25–32. <https://doi.org/10.1016/j.livsci.2018.01.016>.
- Heikkilä, A. M., E. Liski, S. Pyörälä, and S. Taponen. 2018. Pathogen-specific production losses in bovine mastitis. *J. Dairy Sci.* 101:9493–9504. <https://doi.org/10.3168/jds.2018-14824>.

- Hertl, J. A., Y. H. Schukken, F. L. Welcome, L. W. Tauer, and Y. T. Grohn. 2014. Pathogen-specific effects on milk yield in repeated clinical mastitis episodes in Holstein dairy cows. *J. Dairy Sci.* 97:1465–1480. <https://doi.org/10.3168/jds.2013-7266>.
- Kurban, D., Z. Cai, J. P. Roy, T. J. DeVries, J. Ronholm, P. R. F. Adkins, J. R. Middleton, G. P. Keefe, A. France, and S. Dufour. 2025. Dynamics and impacts of staphylococcal intramammary infections during dairy cows' nonlactating periods. *J. Dairy Sci.* 108:11317–11338. <https://doi.org/10.3168/jds.2025-26662>.
- Kurban, D., J. P. Roy, F. Kabera, A. Frechette, M. M. Um, A. Albaaj, S. Rowe, S. Godden, P. R. F. Adkins, J. R. Middleton, M. L. Gauthier, G. P. Keefe, T. J. DeVries, D. F. Kelton, P. Moroni, M. Veiga Dos Santos, H. W. Barkema, and S. Dufour. 2022. Diagnosing intramammary infection: Meta-analysis and mapping review on frequency and udder health relevance of microorganism species isolated from bovine milk samples. *Animals (Basel)* 12:3288. <https://doi.org/10.3390/ani12233288>.
- National Mastitis Council (NMC). 2017. Laboratory Handbook on Bovine Mastitis. 3rd ed. National Mastitis Council, New Prague, MN.
- National Mastitis Council (NMC). 2016. Current Concepts of Bovine Mastitis. 5th ed. National Mastitis Council, New Prague, MN.
- Piepers, S., Y. H. Schukken, P. Passchyn, and S. De Vliegher. 2013. The effect of intramammary infection with coagulase-negative staphylococci in early lactating heifers on milk yield throughout first lactation revisited. *J. Dairy Sci.* 96:5095–5105. <https://doi.org/10.3168/jds.2013-6644>.
- Reyher, K. K., I. R. Dohoo, D. T. Scholl, and G. P. Keefe. 2012. Evaluation of minor pathogen intramammary infection, susceptibility parameters, and somatic cell counts on the development of new intramammary infections with major mastitis pathogens. *J. Dairy Sci.* 95:3766–3780. <https://doi.org/10.3168/jds.2011-5148>.
- Reyher, K. K., S. Dufour, H. W. Barkema, L. Des Côteaux, T. J. DeVries, I. R. Dohoo, G. P. Keefe, J. P. Roy, and D. T. Scholl. 2011. The National Cohort of Dairy Farms—A data collection platform for mastitis research in Canada. *J. Dairy Sci.* 94:1616–1626. <https://doi.org/10.3168/jds.2010-3180>.
- Ruegg, P. 2023. Microbiology, MALDI-TOF, microbiome and more: How to understand and use diagnostic tests to improve udder health. Pages 52–60 in American Association of Bovine Practitioners Conference Proceedings. American Association of Bovine Practitioners.
- Ruegg, P. L. 2017. A 100-Year Review: Mastitis detection, management, and prevention. *J. Dairy Sci.* 100:10381–10397. <https://doi.org/10.3168/jds.2017-13023>.
- Skoulidakis, S., S. Dufour, D. Haine, J. Y. Perreault, and J. P. Roy. 2018. Early-lactation extended pirlimycin therapy against naturally acquired *Staphylococcus aureus* intramammary infections in heifers: A randomized controlled trial. *J. Dairy Sci.* 101:1419–1427. <https://doi.org/10.3168/jds.2017-13726>.
- Supré, K., F. Haesebrouck, R. N. Zadoks, M. Vaneechoutte, S. Piepers, and S. De Vliegher. 2011. Some coagulase-negative *Staphylococcus* species affect udder health more than others. *J. Dairy Sci.* 94:2329–2340. <https://doi.org/10.3168/jds.2010-3741>.
- Taponen, S., J. Koort, J. Björkroth, H. Saloniemi, and S. Pyörälä. 2007. Bovine Intramammary infections caused by coagulase-negative staphylococci may persist throughout lactation according to amplified fragment length polymorphism-based analysis. *J. Dairy Sci.* 90:3301–3307. <https://doi.org/10.3168/jds.2006-860>.
- Taponen, S., V. Myllys, and S. Pyörälä. 2022. Somatic cell count in bovine quarter milk samples culture positive for various *Staphylococcus* species. *Acta Vet. Scand.* 64:32. <https://doi.org/10.1186/s13028-022-00649-8>.
- Taponen, S., and S. Pyörälä. 2009. Coagulase-negative staphylococci as cause of bovine mastitis- not so different from *Staphylococcus aureus*? *Vet. Microbiol.* 134:29–36. <https://doi.org/10.1016/j.vetmic.2008.09.011>.
- Thorberg, B. M., M. L. Danielsson-Tham, U. Emanuelson, and K. Persson Waller. 2009. Bovine subclinical mastitis caused by different types of coagulase-negative staphylococci. *J. Dairy Sci.* 92:4962–4970. <https://doi.org/10.3168/jds.2009-2184>.
- Tomazi, T., J. L. Goncalves, J. R. Barreiro, M. A. Arcari, and M. V. dos Santos. 2015. Bovine subclinical intramammary infection caused by coagulase-negative staphylococci increases somatic cell count but has no effect on milk yield or composition. *J. Dairy Sci.* 98:3071–3078. <https://doi.org/10.3168/jds.2014-8466>.
- Valckenier, D., S. Piepers, A. De Visscher, and S. De Vliegher. 2020. The effect of intramammary infection in early lactation with non-aureus staphylococci in general and *Staphylococcus chromogenes* specifically on quarter milk somatic cell count and quarter milk yield. *J. Dairy Sci.* 103:768–782. <https://doi.org/10.3168/jds.2019-16818>.
- Valckenier, D., S. Piepers, Y. H. Schukken, A. De Visscher, F. Boyen, F. Haesebrouck, and S. De Vliegher. 2021. Longitudinal study on the effects of intramammary infection with non-aureus staphylococci on udder health and milk production in dairy heifers. *J. Dairy Sci.* 104:899–914. <https://doi.org/10.3168/jds.2020-18685>.
- Vanderhaeghen, W., S. Piepers, F. Leroy, E. Van Coillie, F. Haesebrouck, and S. De Vliegher. 2015. Identification, typing, ecology and epidemiology of coagulase negative staphylococci associated with ruminants. *Vet. J.* 203:44–51. <https://doi.org/10.1016/j.tvjl.2014.11.001>.
- Wiggans, G. R., and G. E. Shook. 1987. A lactation measure of somatic cell count. *J. Dairy Sci.* 70:2666–2672. [https://doi.org/10.3168/jds.S0022-0302\(87\)80337-5](https://doi.org/10.3168/jds.S0022-0302(87)80337-5).

ORCIDS

Daryna Kurban, <https://orcid.org/0000-0002-9298-9076>
 Jean-Philippe Roy, <https://orcid.org/0000-0002-0444-2303>
 Trevor J. DeVries, <https://orcid.org/0000-0001-9364-2456>
 Greg P. Keefe, <https://orcid.org/0000-0003-2356-7344>
 Pamela R. F. Adkins, <https://orcid.org/0000-0003-3363-5961>
 John R. Middleton, <https://orcid.org/0000-0003-0791-6604>
 Simon Dufour <https://orcid.org/0000-0001-6418-0424>